Artificial Intelligence

Student Edition

Presented By

LearnKey®

Artificial Intelligence Project Workbook

First Edition

LearnKey

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Introduction

Artificial Intelligence

Best Practices Using LearnKey's Online Training

LearnKey offers video-based training solutions that are flexible enough to accommodate private students and educational facilities and organizations.

Our course content is presented by top experts in their respective fields and provides clear and comprehensive information. The full line of LearnKey products has been extensively reviewed to meet superior quality standards. Our course content has also been endorsed by organizations such as Certiport, CompTIA®, Cisco, and Microsoft. However, it is the testimonials given by countless satisfied customers that truly set us apart as leaders in the information training world.

LearnKey experts are highly qualified professionals who offer years of job and project experience in their subjects. Each expert has been certified at the highest level available for their field of expertise. This expertise provides the student with the knowledge necessary to obtain top-level certifications in their chosen field.

Our accomplished instructors have a rich understanding of the content they present. Effective teaching encompasses presenting the basic principles of a subject and understanding and appreciating organization, real-world application, and links to other related disciplines. Each instructor represents the collective wisdom of their field and within our industry.

Our Instructional Technology

Each course is independently created based on the manufacturer's standard objectives for which the course was developed.

We ensure that the subject matter is up-to-date and relevant. We examine the needs of each student and create training that is both interesting and effective. LearnKey training provides auditory, visual, and kinesthetic learning materials to fit diverse learning styles.

Course Training Model

The course training model allows students to undergo basic training, building upon primary knowledge and concepts to more advanced application and implementation. In this method, students will use the following toolset:

Pre-assessment: The pre-assessment is used to determine the student's prior knowledge of the subject matter. It will also identify a student's strengths and weaknesses, allowing them to focus on the specific subject matter they need to improve the most. Students should not necessarily expect a passing score on the pre-assessment as it is a test of prior knowledge.

Video training sessions: Each training course is divided into sessions or domains and lessons with topics and subtopics. LearnKey recommends incorporating all available external resources into your training, such as student workbooks, glossaries, course support files, and additional customized instructional material. These resources are located in the folder icon at the top of the page.

Exercise labs: Labs are interactive activities that simulate situations presented in the training videos. Step-by-step instructions and live demonstrations are provided.

Post-assessment: The post-assessment is used to determine the student's knowledge gained from interacting with the training. In taking the post-assessment, students should not consult the training or any other materials. A passing score is 80 percent or higher. If the individual does not pass the post-assessment the first time, LearnKey recommends incorporating external resources, such as the workbook and additional customized instructional material.

Workbook: The workbook has various activities, including fill-in-the-blank questions, short answer questions, practice exam questions, and group and individual projects that allow the student to study and apply concepts presented in the course videos.

Using This Workbook

This project workbook contains practice projects and exercises to reinforce the knowledge you have gained through the video portion of the **Artificial Intelligence** course. The purpose of this workbook is twofold. First, to get you further prepared to pass the IT Specialist – Artificial Intelligence exam, and second, to teach you job-ready skills and increase your employability in the area of any job requiring AI model skills.

The projects within this workbook follow the order of the video portion of this course. To save your answers in this workbook, you must first download a copy to your computer. You will not be able to save your answers in the web version. You can complete the workbook exercises as you go through each section of the course, complete several at the end of each domain, or complete them after viewing the entire course. The key is to go through these projects to strengthen your knowledge in this subject.

Each project is based on a specific video (or videos) in the course and specific test objectives. The materials you will need for this course include:

- LearnKey's **Artificial Intelligence** courseware.
- The course project files. All applicable project files are located in the support area where you downloaded this workbook.

For Teachers

LearnKey is proud to provide extra support to instructors upon request. For your benefit as an instructor, we also provide an instructor support .zip file containing answer keys, completed versions of the workbook project files, and other teacher resources. This .zip file is available within your learning platform's admin portal.

Notes

- Extra teacher notes, when applicable, are in the Project Details box within each exercise.
- Exam objectives are aligned with the course objectives listed in each project, and project file names correspond with these numbers.
- The Finished folder in each domain has reference versions of each project. These can help you grade projects.
- Short answers may vary but should be similar to those provided in this workbook.
- Teachers may consider asking students to add their initials, student ID, or other personal identifiers at the end of each saved project.
- Refer to your course representatives for further support.

We value your feedback about our courses. If you have any questions, comments, or concerns, please let us know by visiting https://about.learnkey.com.

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Skills Assessment

Instructions: Rate your skills on the following tasks from 1-5 (1 being needs improvement, 5 being excellent).

Skills	1	2	3	4	5
Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service).	☐	☐	☐	☐	☐
Identify the areas of expertise and resource requirements needed to solve the problem.	☐	☐	☐	☐	☐
Classify the problem (e.g., regression, unsupervised learning)	☐	☐	☐	☐	☐
Plan for AI to be used responsibly.	☐	☐	☐	☐	☐
Choose transparency and validation activities.	☐	☐	☐	☐	☐
Identify data sources and requirements.	☐	☐	☐	☐	☐
Assess data quality.	☐	☐	☐	☐	☐
Convert data into suitable formats (e.g., numerical, image, time series).	☐	☐	☐	☐	☐
Select features for the AI model.	☐	☐	☐	☐	☐
Engage in feature engineering.	☐	☐	☐	☐	☐
Identify training and test data sets.	☐	☐	☐	☐	☐
Document data decisions.	☐	☐	☐	☐	☐
Consider applicability of specific algorithms.	☐	☐	☐	☐	☐
Train a model using the selected algorithm.	☐	☐	☐	☐	☐
Evaluate model accuracy, precision, and sensitivity.	☐	☐	☐	☐	☐
Identify potential sources of bias.	☐	☐	☐	☐	☐
Select specific model after experimentation.	☐	☐	☐	☐	☐
Tell data stories and obtain stakedholder approval.	☐	☐	☐	☐	☐
Design a production pipeline, including application integration.	☐	☐	☐	☐	☐
Identify potential challenges of models in production and plan for AI solution support.	☐	☐	☐	☐	☐

Skills	1	2	3	4	5
Build a security plan.	☐	☐	☐	☐	☐
Train customers on how to use product and what to expect from it.	☐	☐	☐	☐	☐
Engage in oversight.	☐	☐	☐	☐	☐
Assess business impact (key performance indicators).	☐	☐	☐	☐	☐
Measure impacts on individuals and communities.	☐	☐	☐	☐	☐
Handle feedback from users.	☐	☐	☐	☐	☐
Evaluate operations on a regular basis.	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

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Artificial Intelligence Video Times

Domain 1	Video Time
Identify the Problem	00:12:46
Decisions and Benchmarks	00:08:49
Identify Needed Expertise	00:06:27
Identify Resource Requirements	00:06:58
Classify the Problem	00:09:40
Ensure Appropriate Use of AI	00:12:54
Transparency and Validation	00:07:34
Total Time	**01:05:08**

Domain 2	Video Time
Choose the Way to Collect Data	00:07:20
Assess Data Quality	00:10:38
Convert Data	00:07:40
Total Time	**00:25:38**

Domain 3	Video Time
Select Features for the AI Model	00:11:21
Engage in Feature Engineering	00:05:33
Identify Data Sets	00:05:23
Document Data Decisions	00:06:11
Total Time	**00:28:28**

Domain 4	Video Time
Consider Algorithm Applicability	00:06:13
Train a Model	00:04:50
Evaluate Model Performance	00:06:02
Evaluate Model Sensitivity	00:06:54
Look for Potential Bias	00:09:14
Select Specific Model	00:03:23
Tell Data Stories	00:06:18
Obtain Stakeholder Approval	00:03:55
Total Time	**00:46:49**

Domain 5	Video Time
Design a Production Pipeline	00:16:12
Address Potential Challenges	00:17:14
Build a Security Plan	00:05:04
Train Customers on Product	00:06:15
Total Time	**00:44:45**

Domain 6	Video Time
Engage in Oversight	00:11:22
Assess Business Impact	00:05:35
Measure Impacts	00:05:33
Handle User Feedback	00:05:58
Improve or Decommission	00:05:55
Total Time	**00:34:23**

Domain 1
Lesson 1

Artificial Intelligence

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Identify User Persona, Need, Input, and Output

Artificial Intelligence (AI) is a rapidly growing technological concept that is transforming the world of computing. With its various forms and applications across multiple platforms, AI has the power to simplify and complicate people's lives simultaneously. AI's primary purpose is to solve problems, so one ofthe first step in creating an AI project is to identify the need it will address. Once the need is identified, the next step is to determine the input data and the expected output.

Purpose

Upon completing this project, you will become more familiar with AI problem definition, information, and output.

Steps for Completion

1. What is the definition of artificial intelligence?
   a. ____
      ____
      ____
2. List four uses for AI.
   a. ____
      ____
      ____
      ____
3. What are features in an AI project?
   a. ____
4. Determine whether the statement is true or false.
   a. ____________ AI features are always numeric.
5. What is feature engineering, also known as feature selection?
   a. ____
6. Output for an AI model can be a ______________, such as the number of visitors expected on a given day, or a ______________, such as a category people are assigned to for a golf tournament based on their skill levels.
7. What are the five questions that must be answered in order to identify a user persona?
   a. ____
      ____
      ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Identify the Problem
Subtopic: Identify the User Persona, Identify the Need to Be Addressed; Find Out Input and Output

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.1 Identify the user persona – who is the solution for?
1.1.2 Identify the need that will be addressed by the solution
1.1.3 Consider input and output requirements and constraints of the solution

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Determine if AI is Needed

It is essential for a business to assess whether AI is necessary to solve a problem. Evaluating the worth of investing time and resources in AI is crucial. In some cases, like when dealing with historical data results, traditional data analysis methods can be used instead of AI. Identifying how much and what kind of human interaction is needed is also an important step.

Purpose

Upon completing this project, you will better understand how to determine if AI is needed and how much human interaction is required.

Steps for Completion

1. When should an organization consider buying or developing an AI solution?
   a. ______________________________
   ____
   ____
2. Why isn't AI necessary for evaluating historical data results?
   a. ______________________________
3. When data is complex, contains non-linear relationships, or there are large numbers of data, what might be an effective solution?
   a. ______________________________

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Identify the Problem
Subtopic: Determine if AI Is Needed

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.4 Determine whether AI is called for

Domain 1
Lesson 2

Artificial Intelligence

Human Interaction

The next logical step in building an AI solution is to determine whether any interaction between AI and humans is required. Logically, almost every AI solution will require human interaction, so part of this exercise is to determine which parts of the solution do.

Purpose

Upon completing this project, you will better understand how to determine whether human interaction with an AI solution is needed.

Steps for Completion

1. Why is it important to determine which parts of an AI solution require human interaction?
   a. ____
      ____
2. Do the following AI scenarios require human interaction?
   a. __________ Interactions with a chatbot
   b. _________ Park lighting that adjusts based on the color of the sky
   c. __________ Creating staffing schedules

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Decisions and Benchmarks
Subtopic: See if Human Interaction is needed

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.5 Identify whether interaction between AI and humans is required

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Upsides and Downsides of AI

It is essential to understand the advantages and disadvantages of using AI in a situation. On the one hand, AI can provide scalability and availability, but on the other hand, it can potentially lead to job displacement.

Purpose

Upon completing this project, you will become more familiar with the upsides and downsides of AI.

Steps for Completion

1. List three upsides to AI.
    a. ______________________________
    ____
    ____
    ____
2. Determine whether each statement is true or false.
    a. __________ Personal data collected to advertise certain parts of a business to certain demographics is a downside to using AI.
    b. __________ An upside to AI is low costs.
    c. __________ Errors can be made in AI systems due to problems in the algorithms used.
    d. __________ A recreational area can utilize AI to have more efficient staffing numbers in place when needed.
3. Why is it important to understand the upsides and downsides of using AI to solve problems?
    a. ______________________________
    ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Decisions and Benchmarks
Subtopic: Consider Upsides and Downsides

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.6 Consider advantages and disadvantages of using AI in the situation

Define Measurable Success

Defining measurable success is part of solving a problem using AI. The metrics used to measure success depend upon the solution and algorithms used for the solution. There are a few ways in which to evaluate success.

Purpose

Upon completing this project, you will become more familiar with defining measurable success.

Steps for Completion

1. What does AUC-ROC stand for?
   a. ____
   ____

2. Match the metric to its description.

A. Recall	C. AUC-ROC	E. Mean Squared Error (MSE)
B. Accuracy	D. Precision	F. R squared

   a. ________ Percentage of predictions a model gets correct.
   b. ________ A measure of prediction error in numeric situations.
   c. ________ Measures classification at threshold settings.
   d. ________ Measures true positive predictions among all actual positives for classes.
   e. ________ Measures the proportion of variance of a dependent variable that is predictable from independent variables.
   f. ________ Measures percentages of true positive predictions among all positive predictions in an outcome.

3. In the context of the recreational area, the number of visitors is a ____________________ variable, while weather conditions and the day of the week are ____________________ variables.

4. What are two non-statistical measurements of an AI solution?
   a. ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Decisions and Benchmarks
Subtopic: Define Measurable Success

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.7 Define measurable success

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Benchmark Against Risks

When identifying problems that AI can solve, one must consider the organizational risks that could impact any AI project. Certain risks can turn into issues and harm an organization and its reputation.

Purpose

Upon completing this project, you will better understand benchmarking against risks.

Steps for Completion

1. What are the steps for benchmarking against risks?
   a. ______________________________
   b. ______________________________
   c. ______________________________
2. What is a risk mitigation strategy?
   a. ______________________________
3. Determine whether the statement is true or false.
   a. ____________ If an AI project has a lot of high risks, the project can go forward as long as one is aware of them.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Decisions and Benchmarks
Subtopic: Benchmark Against Risks

Objectives covered
1 AI Problem Definition
1.1 Identify the problem we are trying to solve using AI (e.g., user segmentation, improving customer service)
1.1.8 Benchmark against domain or organization-specific risks to which the project may be susceptible

Domain 1
Lesson 3

Artificial Intelligence

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Identify Business and Domain Expertise

To effectively utilize AI to solve a problem, a range of expertise is required, including business acumen. Individuals or a team with an AI project must identify a problem and corresponding solution that warrants the use of AI within an organization. A domain or subject-matter expert should provide guidance and feedback throughout an AI project.

Purpose

Upon completing this project, you will become familiar with identifying business and domain expertise.

Steps for Completion

1. What is the starting point for business expertise when solving a problem using AI?
   a. ____
   ____
   ____
   ____
2. What two types of constraints applying to an AI solution should business experts know?
   a. ____
3. How can a domain (subject-matter) expert help with building an AI solution?
   a. ____
   ____
   ____
4. Determine whether the statement is true or false.
   a. ____________ Even if an organization thinks AI will solve a problem, they should still consult a business expert to ensure invested time and money will pay dividends.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Identify Needed Expertise
Subtopic: Identify Business Expertise; Identify Domain Expertise

Objectives covered
1 AI Problem Definition
1.2 Identify the areas of expertise needed to solve the problem
1.2.1 Identify business expertise required
1.2.2 Identify need for domain (subject-matter) expertise on the problem

Identify AI and Implementation Expertise

AI and implementation expertise focus on the technical needs of an AI solution. Implementation refers to the processes within an AI model used in a solution.

Purpose

Upon completing this project, you will become more familiar with identifying AI and implementation expertise.

Steps for Completion

1. What is AI expertise?
   a. ____________________
   ____
   ____
   ____
2. What are three types of AI solutions?
   a. ____________________
   ____
3. What is an implementation specialist?
   a. ____________________
   ____
   ____
   ____
4. Determine whether each statement is true or false.
   a. __________ Someone without expertise using an AI builder is unlikely to find the right AI solution to solve a problem.
   b. __________ AI solutions can be built using GUI tools, so they do not require knowledge of a programming language.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Identify Needed Expertise
Subtopic: Identify AI Expertise; Identify Implementation Expertise

Objectives covered
1 AI Problem Definition
1.2 Identify the areas of expertise needed to solve the problem
1.2.3 Identify AI expertise needed
1.2.4 Identify implementation expertise needed

Domain 1
Lesson 4

Artificial Intelligence

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Buy or Develop

Developing an AI solution requires extensive programming knowledge in languages such as Python, R, or Java. However, creating an AI solution from scratch is not always necessary. There are many tools available to help build AI solutions. Whether buying or developing an AI model, organizations should consider how the AI model will be stored and what costs are associated with the AI project.

Purpose

Upon completing this project, you will better understand the storage options and costs associated with an AI project.

Steps for Completion

1. When should AI developers consider hosting an AI model locally?
    a. ______________________________
2. When should AI developers consider hosting an AI model in the cloud?
    a. ______________________________
3. List one technical resource that should be considered when budgeting for an AI project.
    a. ______________________________
4. List one human resource that should be considered when budgeting for an AI project.
    a. ______________________________

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Identify Resource Requirements
Subtopic: Assess Existing Resources; Consider Budget and Resources

Objectives covered
1 AI Problem Definition
1.2 Identify the areas of expertise needed to solve the problem
1.2.5 Assess whether problem is solvable. with available computing resources
1.2.6 Consider the budget of the project and resources that are available

Domain 1
Lesson 5

Artificial Intelligence

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Classify the Problem

When there is an AI issue, it is important to identify the type of problem AI will solve and the best course of action for solving it. Data needs to be categorized as labeled or unlabeled. Then, to avoid time wasted on an incorrect solution, the type of problem and the appropriate algorithm to solve the problem need to be identified.

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 1

Topic: Classify the Problem

Subtopic: Examine Available Data; Determine Problem Type

Objectives covered

1 AI Problem Definition

1.3 Classify the problem (e.g., regression, unsupervised learning)

1.3.1 Examine available data

1.3.2 Determine problem type (e.g., classifier, regression, unsupervised, reinforcement)

Purpose

Upon completing this project, you will become more familiar with classifying available data and a problem AI can solve.

Steps for Completion

1. What is the difference between labeled and unlabeled data?
    a. ____
    ____
    ____
2. Match the AI problem category to its definition.

A. Classification	C. Unsupervised
B. Regression	D. Reinforcement

    a. ________ A form of modeling that looks at the relationship between a dependent variable, or target, and an independent variable, or predictor.
    b. ________ A model type that categorizes the result for a given example of input data.
    c. ________ A learning model where an AI agent performs actions to maximize a certain reward.
    d. ________ A learning model that does not look to provide a specific target but is primarily used for finding hidden patterns or structures in data.
3. What is logistic regression?
    a. ____
    ____

Model and Feasibility Needs

When considering an AI solution, a business should assess whether it will need to build a model from scratch. Often, a preexisting model can be used, usually with some fine-tuning. A business should also conduct a feasibility analysis to assess the likelihood of a project being successfully completed.

Purpose

Upon completing this project, you will better understand how to assess model needs and conduct a feasibility analysis.

Steps for Completion

1. What is an AI model?
   a. ____
   ____
   ____
2. Determine whether each statement is True or False.
   a. __________ The existing model that comes with a chatbot will likely need to be fine-tuned.
3. Why might a business need to create a model from scratch?
   a. ____
   ____
4. What are three factors to consider as part of a feasibility analysis?
   a. ____
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Classify the Problem
Subtopic: Examine Model Needs; Conduct Feasibility Needs

Objectives covered
1 AI Problem Definition
1.3 Classify the problem (e.g., regression, unsupervised learning)
1.3.3 Recognize when a pre-trained model is appropriate, if you can fine-tune an existing model, or if you need to train the model from scratch
1.3.4 Conduct feasibility analysis to determine whether it can be realistically modeled

Domain 1
Lesson 6

Artificial Intelligence

Identify Potential User Group Harm

It is important to identify potential ways that AI may mispredict or harm specific user groups when being deployed. To ensure that AI projects do not harm specific user groups, it is important to first understand which groups will be involved in AI projects.

Purpose

Upon completing this project, you will be more familiar with identifying potential user group harm.

Steps for Completion

1. How can one prevent harming specific user groups with AI projects?
   a. ______________________________
   ____
   ____
2. If an AI model for predicting the busiest hours in a park primarily uses data collected during weekdays, what might happen?
   a. ______________________________
   ____
   ____
3. Determine whether each statement is true or false.
   a. ____________ Perfectly avoiding bias is possible in innocuous AI applications.
   b. ____________ An AI system can only postulate about similar samples.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Ensure Appropriate Use of AI
Subtopic: Identify Potential User Group Harm

Objectives covered
1 AI Problem Definition
1.4 Plan for AI to be used responsibly
1.4.1 Identify potential ways that the AI can mispredict and harm specific user groups

Set Data Guidelines

When gathering data to use with an AI model, it is important to ensure that the collected data is both accurate and relevant. One should also ensure that the data does not violate any privacy or security laws of the industry related to the AI model.

Purpose

Upon completing this project, you will better understand setting data guidelines.

Steps for Completion

1. Why should one limit the data users can enter on a user input form?
   a. ____
   ____
2. Determine whether each statement is true or false.
   a. __________ One should assume that all data gathered is not part of the public domain until proven otherwise.
   b. __________ It is not important to cite a researcher when using data from them.
   c. __________ Any personal information that is irrelevant to an AI model being built should be removed.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Ensure Appropriate Use of AI
Subtopic: Set Data Guidelines

Objectives covered
1 AI Problem Definition
1.4 Plan for AI to be used responsibly
1.4.2 Set guidelines for data gathering and use

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Set Algorithm Guidelines

Setting guidelines for algorithm selection from a user perspective, not just a solution developer's, is part of using AI appropriately. Having conversations with the individuals who use the AI system and are looking to improve their processes with it is important.

Purpose

Upon completing this project, you will become more familiar with setting algorithm guidelines.

Steps for Completion

1. Match the user-perspective factors for setting guidelines to their description.

A. Transparency and Explainability	C. Ethics and fairness
B. Empowering users	D. Accessibility

   a. ____________Giving end-users a degree of control over algorithm selection.

   b. ____________Selecting and designing algorithms in a way that promotes fairness and avoids bias.

   c. ____________Favoring models that provide optimal balance between accuracy and interpretability to foster user trust in an AI system.

   d. ____________Providing AI solutions that are easy to use for people with varying degrees of technical proficiency.

2. What are some ways to have conversations with people using an AI system?

   a. ____
   ____

3. What information is needed to determine the group of algorithms to choose from when building an AI model?

   a. ____
   ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Ensure Appropriate Use of AI
Subtopic: Set Algorithm Guidelines

Objectives covered
1 AI Problem Definition
1.4 Plan for AI to be used responsibly
1.4.3 Set guidelines for algorithm selection based on user requirements

Data Interpretation

To create an effective AI solution, it's important not only to choose the most appropriate algorithms but also to consider how data will be presented and interpreted. Different people may prefer different ways of viewing data, such as numerical grids in Excel or charts created in Excel or Power BI.

Purpose

Upon completing this project, you will become more familiar with considering data interpretation.

Steps for Completion

1. Determine whether each statement is true or false.
   a. ________________ It is important to use highly technical terms when explaining AI to users.
   b. _______________ Feedback from people using an AI solution can help with improving those AI products.
2. What does clearly labeling AI-driven results acknowledge?
   a. ____
   ____
3. How can one mitigate the chance of data misinterpretation to foster trust in the AI solution?
   a. ____
4. What can a lack of transparency cause?
   a. ____
   ____
5. What does transparency do?
   a. ____
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Ensure Appropriate Use of AI
Subtopic: Consider Data Interpretation

Objectives covered
1 AI Problem Definition
1.4 Plan for AI to be used responsibly
1.4.4 Consider how the user of the solution can interpret the results

Out-of-Context Use

One aspect of appropriate AI usage is ensuring that the results generated by it are not used out of context. It is important to use the AI solution as intended and within the specific context for which it was designed. Using AI appropriately and responsibly means harnessing its potential benefits while minimizing the risks and potential negative impacts.

Purpose

Upon completing this project, you will better understand how to avoid out-of-context use.

Steps for Completion

1. Why must the scope and limitations of an AI solution be clearly communicated?
   a. ____
   ____
2. Determine whether each statement is true or false.
   a. ____________ One should ensure they communicate the intent of an AI solution to those using it.
   b. ____________ People using an AI solution that works with sensitive personal data must understand the ethical and privacy concerns associated with that data.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Ensure Appropriate Use of AI
Subtopic: Consider Out-of-Context Use

Objectives covered
1 AI Problem Definition
1.4 Plan for AI to be used responsibly
1.4.5 Consider out-of-context use of AI results

Domain 1
Lesson 7

Artificial Intelligence

LearnKey

Transparency in Data Collection and Results

Ensuring transparency in data collection for AI is crucial, as the data often contains personal or sensitive information. Typically, organizations will issue a public announcement outlining the purpose of data collection. Additionally, ensuring transparency and validation activities involves determining who will have access to the results of an AI solution.

Purpose

Upon completing this project, you will become more familiar with communicating the intended purpose of AI data collection and deciding who should see the results.

Steps for Completion

1. What should an organization clearly define in a transparency document?
   a. ______________________________
   ____
2. When deciding who sees AI solution results, what are three things to consider?
   a. ______________________________
   ____
3. What is the principle of least privilege?
   a. ______________________________
   ____
4. What might an organization do when considering regulations on how it handles personal data?
   a. ______________________________
   ____
5. Determine whether each statement is true or false.
   a. __________ When applicable, a transparency document should offer an opt-out clause for anyone who has data that could be collected for an AI system.
   b. __________ People most impacted by an AI project always need to see all the steps required in the process of building the AI model.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Transparency and Validation
Subtopic: Communicate Intended Purpose; Decide Who Should See Results

Objectives covered
1 AI Problem Definition
1.5 Choose transparency and validation activities
1.5.1 Communicate intended purpose of data collection
1.5.2 Decide who should see the results

LearnKey

Review Legal Requirements

Another important transparency and validation activity is reviewing legal requirements specific to an industry as they pertain to AI solutions. AI models are subject to various legal requirements, so AI developers should understand those requirements to ensure any AI model developed remains in compliance with existing legal requirements.

Purpose

Upon completing this project, you will better understand reviewing legal requirements.

Steps for Completion

1. Determine whether each statement is true or false.
    a. _______________ Research is all that is needed to know legal requirements pertaining to AI solutions in an industry.
    b. _______________ An organization may need to conduct privacy impact assessments to discover potential privacy risks and ensure compliance with data protection laws.
2. Other than research, what else can be done so organizations ensure what they are doing complies with regulations?
    a. ____
    ____
    ____
3. What should all efforts taken to reach compliance be? Why?
    a. ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Transparency and Validation
Subtopic: Review Legal Requirement;

Objectives covered
1 AI Problem Definition
1.5 Choose transparency and validation activities
1.5.3 Review legal requirements specific to the industry with the problem being solved

Describe Explainability Methods

Once a business decides to move forward with an AI solution, we want the ability to communicate decisions to stakeholders in a way they can understand. These explanations are crucial, as they can inform decisions about whether to continue the project in progress.

Purpose

Upon completing this project, you will better understand how to explain AI solutions to stakeholders.

Steps for Completion

1. What is a confidence score?
    a. ____
    ____
2. What are two factors in a model, besides the confidence score, that need to be explained to stakeholders?
    a. ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Transparency and Validation
Subtopic: Describe Explainability Methods

Objectives covered
1 AI Problem Definition
1.5 Choose transparency and validation activities
1.5.4 Describe explainability methods for communicating AI decisions to stakeholders.

Domain 2
Lesson 1

Artificial Intelligence

LearnKey

Data Collection

When working with data for an AI model, one must determine the type and characteristics needed to use AI to solve a problem. Once data types and characteristics are defined, the next step in the data collection process is to determine whether an existing data set is a good starting point for an AI model or whether data needs to be generated.

Purpose

Upon completing this project, you will better understand the data collection process for an AI model.

Steps for Completion

1. Determine whether each statement is true or false.
    a. _______________ The more complex an AI problem is, the more data is needed.
    b. _______________ A data set must be small to produce accurate results.
    c. _______________ Users should not sacrifice the quality of data to save time.
    d. _______________ Automated data processes only collect what they are programmed to collect.
    e. _______________ Data can only be collected from existing data sets.
2. Explain why data must be balanced.
    a. ____
    ____
    ____
    ____
    ____
3. List three data collection methods.
    a. ____
    ____
4. Describe one example of an AI problem that is well-suited for automatically collected data.
    a. ____
    ____
5. Describe one example of an AI problem that is well-suited for data collected from user input.
    a. ____
    ____

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 2
Topic: Choose the Way to Collect Data
Subtopic: Determine Data Types Needed; Decide if Needed Data Exists; Choose Data collection Methods, Automated vs. User Input

Objectives covered
2 Data Collection and Transformation
2.1 Identify data sources and requirements
2.1.1 Determine type/characteristics of data needed
2.1.2 Decide if there is an existing data set or if you need to generate your own
2.1.3 Choose data collection methods
2.1.4 When generating your own data set, decide whether collection can be automated or requires user input

Domain 2
Lesson 2

Artificial Intelligence

LearnKey

Clean Data

After AI developers determine that the data they have collected will meet the needs of the AI project they are working on, the collected data must be cleaned. Cleaning data involves identifying missing, misaligned, fake, and corrupted data and then implementing techniques that help prevent flawed data from skewing AI results.

Purpose

Upon completing this project, you will better understand how to clean collected data.

Steps for Completion

1. How can users handle missing data?
    a. ______________________________
2. How can users handle misaligned data?
    a. ______________________________
3. How can users handle fake or corrupt data?
    a. ______________________________

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Assess Data Quality
Subtopic: Look for Missing or Corrupt Data

Objectives covered
2 Data Collection and Transformation
2.2 Assess data quality
2.2.1 Identify missing or corrupt data elements

LearnKey

Bias in Data

AI developers must ensure that the data they collect represents what they are trying to accomplish. One of the biggest detriments to having quality data is bias. Data collection tools can be prone to bias in different ways.

Purpose

Upon completing this project, you will better understand how to evaluate data for bias.

Steps for Completion

1. What is selection bias?
   a. ______________________________
   ____
   ____
2. What is historical bias?
   a. ______________________________
   ____
   ____
3. Describe one way in which survey tools can be biased.
   a. ______________________________
   ____
   ____
   ____
4. Why might online data collection tools be biased?
   a. ______________________________
   ____
   ____
5. Describe one way in which observational data collection tools can be biased.
   a. ______________________________
   ____
   ____
6. What can AI developers do if they determine a data set is too small?
   a. ______________________________
7. A data set should be large enough to produce a(n) ____________________ AI model.
8. How can incorrect annotations cause bias?
   a. ______________________________
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Ensure Representation
Subtopic: Examine Collection Techniques; Ensure Enough Data is Unbiased; Assess the Quality of the Annotation

Objectives covered
2 Data Collection and Transformation
2.2 Assess data quality
2.2.2 Examine collection techniques for potential sources of bias
2.2.3 Make sure the amount of data is enough to build an unbiased model
2.2.4 Assess the quality of the annotation

Domain 2
Lesson 3

Artificial Intelligence

LearnKey

Convert Images to Binary Format

In many AI systems, data needs to be converted from whatever format it arrives into binary, such as zeros and ones, to be useful for an AI model. This conversion especially applies to images. Images must be converted to binary format for an AI model to be able to process and use the data from the images in results.

Purpose

Upon completing this project, you will better understand how to convert images to binary format.

Steps for Completion

1. What does it mean to transform an image's data into binary format?
    a. ____
    ____
    ____
    ____
2. What is one-hot encoding?
    a. ____
3. What are word embeddings?
    a. ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Convert Data
Subtopic: Convert to Binary

Objectives covered
2 Data Collection and Transformation
2.3 Convert data into suitable formats (e.g., numerical, image, time series)
2.3.1 Convert data to binary (e.g., images become pixels)

Convert Sentences to Tokens

In addition to converting images to binary format, AI developers may need to convert text to tokens. Tokens are smaller units of words or sentences. AI developers should remember that to process data and make predictions, an AI model needs to convert data, whether from images or text, into a language it can understand to properly evaluate data.

Purpose

Upon completing this project, you will better understand how to convert sentences to tokens.

Steps for Completion

1. List the steps of converting sentences to tokens in the correct order.
    a. ______________________________
    b. ______________________________
    c. ______________________________
    d. ______________________________
2. Why might AI developers convert sentences to tokens?
    a. ______________________________
    ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Convert Data
Subtopic: Convert to Features Suitable for AI

Objectives covered
2 Data Collection and Transformation
2.3 Convert data into suitable formats (e.g., numerical, image, time series)
2.3.2 Convert computer data into features suitable for AI (e.g., sentences become tokens)

LearnKey

Data Annotation

Once data has been collected, one needs to determine whether it is necessary to annotate the data. Proper annotation when the situation calls for it is key to ensuring an AI model is trained properly. However, this process is not necessary for all data, such as when the AI model is trying to find patterns on its own.

Purpose

Upon completing this project, you will better understand how to identify data that needs to be annotated.

Steps for Completion

1. What is an annotation?
   a. ______________________________
   ____
2. List three scenarios that require data to be annotated.
   a. ______________________________
   ____
3. Describe a situation that does not require data to be annotated.
   a. ______________________________
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Convert Data
Subtopic: Annotate the Data if Necessary

Objectives covered
2 Data Collection and Transformation
2.3 Convert data into suitable formats (e.g., numerical, image, time series)
2.3.3 Annotate the data if necessary

Domain 3
Lesson 1

Artificial Intelligence

Features

A feature is a characteristic used in AI machine learning. An AI model using incorrect features will not produce a usable outcome, thus wasting time and money in an organization's quest to make AI work for it. Including only features that belong can help ensure an AI model performs well and has likely results.

Purpose

Upon completing this project, you will better understand how to determine what features to include in an AI model.

Steps for Completion

1. What two questions should AI developers ask when determining what features to include in an AI model?
   a. ______________________________
   ____
   b. ______________________________
   ____
2. How can a correlation between two data points be used to determine whether a feature should be included in an AI model?
   a. ______________________________
   ____
   ____
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Select Features for the AI Model
Subtopic: Determine Features to Include

Objectives covered
3 Feature Engineering
3.1 Select features for the AI model
3.1.1 Determine which data features to include

LearnKey

Feature Vectors

In addition to selecting appropriate features for a testing or training data set for an AI model, turning some of those features into feature vectors may also be necessary. Building initial feature vector data sets to train or test an AI model is a crucial step in developing AI solutions. Following the process for developing initial feature vectors is essential to developing robust and reliable AI models.

Purpose

Upon completing this project, you will better understand what feature vectors are and why they are important for an AI model.

Steps for Completion

1. What is a feature vector?
   a. ______________________________
   ____
   ____
   ____
   ____
2. Why are feature vectors needed when developing an AI model?
   a. ______________________________
   ____
   ____
3. What does it mean to scale feature vectors?
   a. ______________________________
   ____
4. What is encoding?
   a. ______________________________
5. Determine whether each statement is true or false.
   a. __________ An AI model's performance on random data or a small, unrepresentative data set may not be indicative of its performance on real-world data.
   b. __________ The data the model has access to usually comes in two forms: the training data set and the archived data set.
   c. __________ It is crucial that the testing data is mixed with the training data to get an unbiased estimate of an AI model's performance.
   d. __________ In many instances, AI developers will consult with subject matter experts to help ensure appropriate features are used in an AI model.
   e. __________ AI developers should not consult subject matter experts about risky features.

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 3
Topic: Select Features for the AI Model
Subtopic: Build Initial Feature Vectors; Consult with SMEs on Selection

Objectives covered
3 Feature Engineering
3.1 Select features for the AI model
3.1.2 Build initial feature vectors for test/train data set
3.1.3 Consult with subject-matter experts to confirm feature selection

Domain 3
Lesson 2

Artificial Intelligence

LearnKey

Transform and Manipulate Data

Once features are selected for an AI model, AI developers should engage in feature engineering to ensure the features work properly within the context of the model. A data set used for AI is rarely ready for an AI model without data transformation and manipulation.

Purpose

Upon completing this project, you will better understand how to transform and manipulate data.

Steps for Completion

1. What occurs during the process of feature engineering?
   a. ____
   ____
2. What three methods can AI developers use to transform and manipulate data to prepare it for an AI model?
   a. ____
   ____
   ____
3. Determine whether each statement is true or false.
   a. ______________ Once data is transformed, AI developers should validate the transformation.
   b. ______________ If missing values in data sets are critical for an AI model to be accurate, AI developers may consider estimating the values and entering them into the model.
   c. ______________ AI developers can add keywords to mark existing words in data that help categorize data for a model.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Engage in Feature Engineering
Subtopic: Determine Needed Transformations; Create Processed Datasets; Identify Transformation Application

Objectives covered
3 Feature Engineering
3.2 Engage in feature engineering
3.2.1 Review features and determine what standard transformations are needed
3.2.2 Create processed data sets
3.2.3 Identify whether transformations have been applied correctly

Domain 3
Lesson 3

Artificial Intelligence

LearnKey

Training and Test Data Sets

Once we have a data set and decide what features we will use in an AI model, the next step is to separate the available data into two sets: training and test sets. Test sets are also known as validation sets.

Purpose

Upon completing this project, you will better understand the purpose of using training and test data sets on an AI model.

Steps for Completion

1. What is a test data set used for?
   a. ____
   ____
2. What is a training data set used for?
   a. ____
   ____
3. What is the goal of using test data sets and training data sets on an AI model?
   a. ____
   ____
   ____
   ____
4. How can AI developers accurately evaluate a model's performance using test data sets?
   a. ____
   ____
5. Describe two ways in which random sampling can be beneficial to creating accurate test data sets.
   a. ____

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 3
Topic: Identify Data Sets
Subtopic: Training and Test Sets; Ensure Test Set is Representative

Objectives covered
3 Feature Engineering
3.3 Identify training and test data sets
3.3.1 Separate available data into training and test sets
3.3.2 Ensure test set is representative

Domain 3
Lesson 4

Artificial Intelligence

LearnKey

Documentation

There are three areas of importance in documenting data decisions: listing assumptions, documenting predicates, and specifying constraints. These three areas impact not only the data decisions made within documents but also how AI models are built to ensure that the models fit within the assumptions and constraints associated with AI projects. Creating and maintaining white papers is crucial for effective documentation of AI projects.

Purpose

Upon completing this project, you will better understand how to document various aspects of AI models.

Steps for Completion

1. What are assumptions?
   a. ____
   ____
   ____
2. What are predicates?
   a. ____
   ____
   ____
3. What are constraints?
   a. ____
   ____
   ____
4. What is the purpose of white papers?
   a. ____
   ____
5. Where are two places white papers should be stored?
   a. ____
   ____
6. What is deep transparency?
   a. ____
   ____
7. What documents should be created while planning an AI project?
   a. ____
8. List two documents needed while an AI model is being built.
   a. ____

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 3
Topic: Document Data Decisions
Subtopic: Assumptions, Predicates, and Constraints; Make Information Available; Identify Relevant Documentation

Objectives covered
3 Feature Engineering
3.4 Document data decisions
3.4.1 List assumptions, predicates, and constraints upon which design choices have been reasoned
3.4.2 Make this information available to regulators and end users who demand deep transparency
3.4.3 Identify situations in which particular types of documentation are relevant

Domain 4
Lesson 1

Artificial Intelligence

AI Algorithm Families

In AI, an algorithm processes data in a model to produce a prediction or outcome. There are several prominent families of algorithms to know when choosing one for an AI model: supervised learning, unsupervised learning, reinforcement learning, Natural Language Processing (NLP), and regression. Knowing these algorithm families helps organizations choose the correct algorithms for their AI projects.

Purpose

Upon completing this project, you will become more familiar with AI algorithm families.

Steps for Completion

1. What is an algorithm?
   a. ____
2. Match the algorithm family to its description.

A. Supervised learning	D. Reinforcement learning
B. Unsupervised learning	E. Regression
C. Natural Language Processing	

   a. ________ Algorithms that do not work with labeled data, meaning the outcomes or groups of outcomes are somewhat unknown.
   b. ________ Algorithms that are often used for reward systems and automatic game-playing. They often learn from interacting with their environment.
   c. ________ Algorithms that use labeled data, meaning data where a target variable is known.
   d. ________ Algorithms used to predict numbers.
   e. ________ Algorithms that interpret or respond to human text or speech.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Consider Algorithm Applicability
Subtopic: Evaluate AI Algorithm Families; NLP and Regression

Objectives covered
4 AI Algorithms and Models
4.1 Consider applicability of specific algorithms
4.1.1 Evaluate AI algorithm families

Decide on Suitable Algorithms

It is essential to understand the applicability of AI algorithms to determine which ones are appropriate for a project. A clear understanding of these algorithms is crucial in selecting the most suitable one for a particular scenario, thus producing the best possible output for solving an AI problem.

Purpose

Upon completing this project, you will better understand how to decide on suitable algorithms.

Steps for Completion

1. What is a neural network algorithm?
    a. ______________________________
    ____
    ____
2. Which algorithm is a supervised learning method used mainly for classification?
    a. ______________________________
3. What is k-means clustering helpful for?
    a. ______________________________
    ____
4. What are the two most significant considerations when deciding which algorithm to use?
    a. ______________________________
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Consider Algorithm Applicability
Subtopic: Decide on Suitable Algorithms

Objectives covered
4 AI Algorithms and Models
4.1 Consider applicability of specific algorithms
4.1.2 Decide which algorithms are suitable, e.g., neural network, classification (like decision tree, k-means)

Domain 4
Lesson 2

Artificial Intelligence

LearnKey

Train with Parameters

After selecting an algorithm, the next step in an AI project is to train a model with best-guess starting parameters. The training process should be documented, and any parameter changes should be noted.

Purpose

Upon completing this project, you will become more familiar with training a model with parameters.

Steps for Completion

1. Once training data is cleansed and transformed, what happens?
   a. ____
   ____
   ____
2. Why should the training process be documented?
   a. ____
   ____
   ____
3. Determine whether each statement is true or false.
   a. ______________ Training through multiple iterations may be necessary to produce the desired output for an AI project.
   b. ______________ If the evaluation results of a model's performance are not as anticipated, a new model must be built.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Train a Model
Subtopic: Train with Parameters

Objectives covered
4 AI Algorithms and Models
4.2 Train a model using the selected algorithm
4.2.1 Train model using an algorithm with starting parameters

LearnKey

Train a Model

When training an AI model, it is essential to understand which parameters can be modified to fine-tune the model and when and why they should be adjusted. Additionally, it is necessary to collect performance metrics at some point during the training process to ensure that the model is functioning as expected. By making iterative adjustments to the model using different settings, such as other parameters, it is possible to improve its performance.

Purpose

Upon completing this project, you will become more familiar with changing parameters and gathering performance metrics to tune a model.

Steps for Completion

1. Determine whether each statement is true or false.
    a. ________________ A model can be trained using multiple iterations of training, each with a different parameter combination.
    b. ________________ To best evaluate a model, one should use the same data set used for training.
    c. ________________ Identifying areas for improvement within a model starts with analyzing the model's performance to identify weaknesses.
    d. ________________ Parameters can be changed to remove all bias in an AI model's data.
2. What are three common metrics?
    a. ____
    ____
    ____
3. Why is it important for iterative adjustments to be well-documented?
    a. ____
    ____
    ____
4. What is a test or validation data set?
    a. ____
    ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Train a Model
Subtopic: Tune by Changing Parameters; Gather Performance Metrics; Iterate as Needed

Objectives covered
4 AI Algorithms and Models
4.2 Train a model using the selected algorithm
4.2.2 Tune the model by changing parameters
4.2.3 Identify performance metrics for the model
4.2.4 Determine whether additional iterations are required

Domain 4
Lesson 3

Artificial Intelligence

LearnKey

Overfitting and Underfitting

Accuracy and precision are crucial factors for ensuring an AI model performs as it should. Two concepts related to these factors are overfitting and underfitting. When overfitting or underfitting occurs, it is important to address these issues to ensure that the model works well on both trained and unseen data.

Purpose

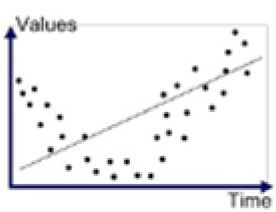
Upon completing this project, you will become more familiar with overfitting and underfitting.

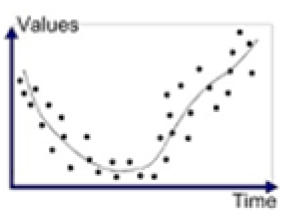
Steps for Completion

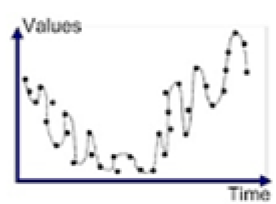
1. What is overfitting?
   a. ____
   ____
   ____
   ____

2. What is underfitting?
   a. ____
   ____
   ____
   ____

3. Match the chart to its description.

A. 

B. 

C. 

   a. __________ Good fit
   b. __________ Underfitted
   c. __________ Overfitted

4. What are two ways to mitigate underfitting and overfitting?
   a. ____
   ____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 3
Topic: Evaluate Model Performance
Subtopic: Check for Overfitting and Underfitting

Objectives covered
4 AI Algorithms and Models
4.3 Evaluate model accuracy, precision, and sensitivity
4.3.1 Check for overfitting, underfitting

LearnKey

Generate Metrics or KPIs

When evaluating an AI model, it is important to generate metrics or key performance indicators (KPIs) to determine whether the model performs as expected. To generate these metrics, one must establish an acceptable threshold for accuracy based on the industry and other relevant factors.

Purpose

Upon completing this project, you will better understand generating metrics or KPIs and their role in determining an AI model's performance.

Steps for Completion

1. What are KPIs?
   a. ______________________________
   ____
   ____
2. What is the purpose of a confusion matrix?
   a. ______________________________
   ____
   ____
3. Match the metric to its formula.

A. Precision	C. Accuracy
B. Recall	D. F1 Score

   a. ________ True Positives / (True Positives + False Negatives)
   b. ________ True Positives / (True Positives + False Positives)
   c. ________ 2 x Precision x Recall / Precision + Recall
   d. ________ (True Positives + True Negatives) / Total Predictions
4. Determine whether the statement is true or false.
   a. ______________ A confusion matrix with a high accuracy percentage but a very low number of true positives or negatives is just as reliable as one with a slightly lower accuracy percentage but a healthy number of true positives and true negatives.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Evaluate Model Performance
Subtopic: Generate Metrics or KPIs

Objectives covered
4 AI Algorithms and Models
4.3 Evaluate model accuracy, precision, and sensitivity
4.3.2 Generate metrics or KPIs (accuracy, precision, recall, F1 score, etc.)

Introduce New Test Data

While an AI model must work well with test data, it also must work well with new, unforeseen data. To evaluate a model's performance with unforeseen data, one must ensure that the new test data is distinct from any previous training or testing iterations for the model.

Purpose

Upon completing this project, you will become more familiar with introducing new test data to an AI model.

Steps for Completion

1. What three things must be done with new data before testing it?
   a. ______________________________
   ____
   ____
2. Why is it important to document all the steps to prepare new test data?
   a. ______________________________
   ____
   ____
3. Why should one document and communicate changes to important stakeholders?
   a. ______________________________
4. Determine whether the statement is true or false.
   a. __________ If new data introduces possible bias into a model, the size of the data set, the data itself, or the model itself may need to be adjusted.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Evaluate Model Performance
Subtopic: Introduce New Test Data

Objectives covered
4 AI Algorithms and Models
4.3 Evaluate model accuracy, precision, and sensitivity
4.3.3 Introduce new test data to cross-validate robustness, testing how model handles unforeseen data

Domain 4
Lesson 4

Artificial Intelligence

LearnKey

Test for Model Sensitivity and Specificity

Two important measures for assessing the performance of an AI model are sensitivity and specificity tests. One should consider both measures when building and evaluating AI models.

Purpose

Upon completing this project, you will become more familiar with testing for model sensitivity.

Steps for Completion

1. What is sensitivity in the context of an AI model?
   a. ______________________________
   ____
   ____
2. What is specificity in the context of an AI model?
   a. ______________________________
   ____
   ____
3. If a small change in a model's input data leads to a significant change in the output, what does it mean?
   a. ______________________________
4. Sensitivity refers to how well a model identifies ____________ instances, while specificity refers to how well a model identifies ____________ instances.
5. Determine whether each statement is true or false.
   a. ____________ While testing a recreational area model, if changing weather conditions do not affect the number of visitors, the model is highly sensitive to weather conditions.
   b. ____________ While testing a recreation area model, if changing wind direction does not affect the number of visitors, the model has low sensitivity to wind direction.
   c. ____________ If a model is good at recognizing when a situation meets the criteria for a positive result, it has high specificity.
   d. ____________ To test the specificity of a model, one would need a data set that includes instances where alerts should not be triggered and check if the model correctly identifies these as non-alert situations when conditions can trigger alerts.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Evaluate Model Sensitivity
Subtopic: Test for Model Sensitivity; Test for Model Specificity

Objectives covered
4 AI Algorithms and Models
4.3 Evaluate model accuracy, precision, and sensitivity
4.3.4 Interpret performance metrics (sensitivity, specificity, confusion matrix)

Model Improvement Techniques

After evaluating model performance, we will want to improve it. There are several techniques we can use to improve model performance. It is important to know which techniques to use and when, so our AI models can generate data and accurate predictions when needed.

Purpose

Upon completing this project, you will better understand the techniques available to improve model performance.

Steps for Completion

1. How can more data improve model performance?
    a. ____
    ____
    ____
2. Determine whether the following statements are true or false.
    a. __________ Training data should try to obtain the same data splits as in the overall data sets being used.
    b. __________ Cleaning data will not typically impact a model's performance.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Evaluate Model Sensitivity
Subtopic: Model Performance Improvement Techniques

Objectives covered
4 AI Algorithms and Models
4.3 Evaluate model accuracy, precision, and sensitivity
4.3.5 Identify appropriate techniques to improve model performance

Domain 4
Lesson 5

Artificial Intelligence

LearnKey

Verify Inputs Resemble Training Data

One of the biggest challenges facing AI is the issue of bias. Bias can creep into AI models in several ways, such as through the data used to train the models or the algorithms used to analyze the data. To ensure the accuracy and fairness of AI, it is essential that data used in AI models is free from bias or that steps are taken to mitigate any bias that may be present.

Purpose

Upon completing this project, you will become more familiar with verifying that inputs resemble training data.

Steps for Completion

1. What is the purpose of comparing training data to real-world inputs?
    a. ______________________________
2. What are three ways in which data can be biased?
    a. ______________________________
3. Determine whether the statement is true or false.
    a. __________ If training data is unbiased, real-world data input after the fact will be unbiased.
    b. __________ Mitigating bias in real-world data means making changes to data-collecting methods.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Look for Potential Bias
Subtopic: Verify Inputs Resemble Training Data

Objectives covered
4 AI Algorithms and Models
4.4 Identify potential sources of bias
4.4.1 Verify that inputs resemble training data

LearnKey

Irrelevant Correlations and Data Imbalances

One method for mitigating bias in AI models is to eliminate irrelevant correlations from the data set. One of the main causes of bias in AI models is data imbalance, particularly concerning demographics.

Purpose

Upon completing this project, you will become more familiar with irrelevant correlations and data imbalances.

Steps for Completion

1. Irrelevant correlations make classifications unreliable in AI models, leading to what?
   a. ________________________________
2. Which function in Excel can be used to find correlations in data?
   a. ________________________________
3. Which data type is it important to pay particular attention to regarding imbalances?
   a. ________________________________
4. Open the **362-boaters.xlsx** file from your Domain 3 Student folder.
5. A correlation is shown between the boaters and which data category in the spreadsheet? How do you know?
   a. ________________________________
   b. ________________________________

Project Details

Project file
362-boaters.xlsx

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Look for Potential Bias
Subtopic: Confirm Correlations; Check for Data Imbalances

Objectives covered
4 AI Algorithms and Models
4.4 Identify potential sources of bias
4.4.2 Confirm that training data do not contain irrelevant correlations we do not want classifier to rely on
4.4.3 Check for imbalances in data

LearnKey

Guard Against Self-Fulfilling Prophecies

Data bias can take many forms, including the self-fulfilling prophecy. In AI, this type of bias occurs when an algorithm reinforces and amplifies existing patterns in data, resulting in biased outcomes that perpetuate the biases present in the original data, the people developing the AI model, or one's personal expected results of an AI model.

Purpose

Upon completing this project, you will better understand how to guard against self-fulfilling prophecies.

Steps for Completion

1. What is historical bias?
    a. ______________________________
    ____
    ____
2. An AI model predicts visitor numbers for a park and influences the park's advertising budget. In turn, the advertising influences visitor numbers for the park. What has been created?
    a. ______________________________
3. What can including counterfactual reasoning in an algorithm's design help correct?
    a. ______________________________
    ____
4. Determine whether the statement is true or false.
    a. __________ The goal should always be to create an AI model that provides accurate and unbiased predictions.
5. How might one mitigate the issues of self-fulfilling prophecies?
    a. ______________________________
    ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Look for Potential Bias
Subtopic: Guard Against Self-Fulfilling Prophecies

Objectives covered
4 AI Algorithms and Models
4.4 Identify potential sources of bias
4.4.4 Guard against creating self-fulfilling prophecies based upon historical biases

Domain 4
Lesson 6

Artificial Intelligence

Cost, Speed, Explainability, and Other Factors

There are several factors to consider when evaluating models for an AI project. These factors include cost, speed, explainability, and others. Evaluating models based on these factors helps ensure AI models serve their intended purposes and warrant the time and money invested. Ensuring a model meets explainability requirements is essential, as it promotes transparency and trust in AI systems.

Purpose

Upon completing this project, you will better understand cost, speed, explainability, and other factors when evaluating AI models.

Steps for Completion

1. How might cost factor into developing an AI model?
   a. ____
   ____
   ____
2. For which type of app is speed a more significant factor?
   a. ____
   ____
   ____
3. What are explainability requirements?
   a. ____
   ____
4. Why is explainability especially important in a field such as healthcare?
   a. ____
   ____
5. Determine whether each statement is true or false.
   a. ______________ AI models are used to predict known outcomes.
   b. ______________ If a model is sluggish, it will not work well in real-time situations.
   c. ______________ Part of explainability includes ensuring an AI model complies with regulatory or legal requirements for the data it is using.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Select Specific Model
Subtopic: Cost, Speed, and Other Factors; Determine if Model Meets Explainability

Objectives covered
4 AI Algorithms and Models
4.5 Select specific model after experimentation
4.5.1 Consider cost, speed, and other factors in evaluating models
4.5.2 Determine whether the selected model meets explainability requirements

Domain 4
Lesson 7

Artificial Intelligence

Identify Visualization Types

One way to explain the results of an AI model is to create visualizations of the results. Visualizations include charts, reports, and similar tools for summarizing data and presenting it in a way understandable to most, if not all, audiences that will benefit from an AI model. For visualizations to be effective, the right type is needed for the data being reported.

Purpose

Upon completing this project, you will better understand how to identify which visualization types work best for which types of data.

Steps for Completion

1. Match the type of visualization to its description.

A. Line chart	D. Scatter plot
B. Bar chart	E. Histogram
C. Pie chart	

   a. ________ A chart that represents the distribution of numerical data.

   b. ________ A chart that compares items as they make up a whole.

   c. ________ A chart that reports trends over time.

   d. ________ A chart that compares multiple series.

   e. ________ A chart that shows the relationship between two data sets.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Tell Data Stories
Subtopic: Identify Visualization Types

Objectives covered
4 AI Algorithms and Models
4.6 Tell data stories and obtain stakeholder approval
4.6.1 Identify which visualization types will represent the data story most appropriately

Create Visualizations

Visualizing the results of an AI model is an effective method of explaining its performance. Creating visual representations makes it easier to determine a model's accuracy and identify areas that require further training. This iterative process of analyzing and refining the model can significantly improve its overall performance.

Purpose

Upon completing this project, you will become more familiar with creating visualizations.

Steps for Completion

1. What are visualizations?
    a. ____
    ____
    ____
2. What can using visualizations to show training model data help identify?
    a. ____
    ____
    ____
3. Determine whether each statement is true or false.
    a. __________ When a chart used to visualize data is difficult to understand, a different chart can be used.
    b. __________ Visualizations should only be used to show predictions for a completed AI model.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Tell Data Stories
Subtopic: Create Visualizations

Objectives covered
4 AI Algorithms and Models
4.6 Tell data stories and obtain stakeholder approval
4.6.2 Where feasible, create visualizations of the results

LearnKey

Look for Trends and Verify Visualizations

It is crucial to search for trends in AI models to ensure the accuracy of the data used. Identifying trends also ensures one is better informed when making business decisions based on AI models. After creating visualizations to explain AI model results, one must verify that these visualizations help make informed business decisions.

Purpose

Upon completing this project, you will become more familiar with looking for trends and verifying visualizations.

Steps for Completion

1. What is often the reason for seeing one or more outliers in data that do not follow a trend?
   a. ____
   ____
2. Determine whether each statement is true or false.
   a. ____________ An overarching theme of AI is that results must make sense.
   b. ____________ Trends in data cannot be identified without a chart.
   c. ____________ Visualizations should help determine whether an AI model is accurate.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Tell Data Stories
Subtopic: Look for Trends; Verify That the Visualization Is Useful

Objectives covered
4 AI Algorithms and Models
4.6 Tell data stories and obtain stakeholder approval
4.6.3 Look for trends
4.6.4 Verify that the visualizations are useful for making a decision

Domain 4
Lesson 8

Artificial Intelligence

Results, Risks, and Evaluation Sessions

Collecting results and benchmarking risks helps make an AI model easier for stakeholders to understand. Stakeholders often want to understand an AI project before giving their approval to move forward with it. Holding session meetings to evaluate an AI solution may also be necessary for obtaining stakeholder approval.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 4

Topic: Obtain Stakeholder Approval

Subtopic: Collect Results and Benchmark Risks; Hold Evaluation Sessions

Objectives covered

4 AI Algorithms and Models

4.6 Tell data stories and obtain stakeholder approval

4.6.5 Collect results and benchmark risks and hold sessions to evaluate the solution

Purpose

Upon completing this project, you will become more familiar with collecting results and benchmarking risks.

Steps for Completion

1. What is a stakeholder?
   a. ____
   ____
   ____
2. What should a solution outcome be compared to when helping stakeholders understand data?
   a. ____.
3. Determine whether each statement is true or false.
   a. ____________ Customers and employees are examples of stakeholders for the recreational area AI projects.
   b. ____________ Subject matter experts, users, and legal representatives are all potential stakeholders for an AI solution.
   c. ____________ All stakeholders are interested in the exact algorithms used to generate an AI solution.
   d. ____________ Legal representatives are not potential stakeholders for an AI solution.

Domain 5
Lesson 1

Artificial Intelligence

LearnKey

AI Model Pipeline

After evaluating the components and challenges of an AI model, the next step in the AI process is to build a model pipeline. When creating an AI model pipeline, AI developers will have existing data stores containing data that they can use to build a pipeline. To create a well-working AI model pipeline, AI developers must address the integration and transformation requirements of the data that will be used in the AI model.

Purpose

Upon completing this project, you will better understand how to prepare data for an AI model pipeline.

Steps for Completion

1. What is the purpose of an AI model pipeline?
   a. ______________________________
   ____
2. Describe the process of continuous integration and deployment (CI/CD).
   a. ______________________________
   ____
   ____
   ____
3. What is the 10 rule?
   a. ______________________________
   ____
4. What four steps can AI developers use to help ensure data works well with an AI pipeline?
   a. ______________________________
   b. ______________________________
   c. ______________________________
   d. ______________________________

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Design a Production Pipeline
Subtopic: Plan for CI/CD; Create a Pipeline; Find a Solution

Objectives covered
5 Application Integrity and Deployment
5.1 Design a production pipeline, including application integration
5.1.1 Plan for continuous itegration and deployment (CI/CD)
5.1.2 Design a pipeline that integrates with existing data stores and connects to the application

LearnKey

Using a Pipeline

As part of building a pipeline for an AI model, AI developers must build one or more connections between AI and the applications with the data being used to develop the AI model. To do this effectively, AI developers first need to make sure they understand any integration requirements going from the model to any data that it accesses. As part of building and testing an AI model, AI developers must build at least one mechanism to gather user feedback when deploying an application.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 5
Topic: Design a Production Pipeline
Subtopic: Build App Connections; Build User Feedback Mechanism

Objectives covered
5 Application Integrity and Deployment
5.1 Design a production pipeline, including application integration
5.1.3 Build the connection between the AI and the application

Purpose

Upon completing this project, you will better understand how to build an AI model pipeline.

Steps for Completion

3. What are two ways AI developers can make connections between an AI model and data?
    a. ______________________________
    ____
4. What is the purpose of error and exception handling?
    a. ______________________________
    ____
    ____
5. What is sentiment analysis?
    a. ______________________________
    ____
6. Determine whether each statement is true or false.
    a. ____________ All data connections require code.
    b. ____________ The requirements for taking data from an outside source and transforming it for an AI model should be outlined after the AI model pipeline is created.
    c. ____________ Once an AI model is deployed, AI developers should monitor those connections for possible interruptions.

LearnKey

Testing AI Models

Testing the accuracy of AI models helps ensure those models produce relevant and applicable results. Testing for robustness helps ensure that AI systems work well from a data standpoint and a functional standpoint, especially in peak periods of usage. AI developers should test the speed of any AI system, as a slow system will become unpopular and not be used to its full capabilities. AI applications should also fit their intended use cases, such as being used on mobile devices.

Purpose

Upon completing this project, you will better understand how to test an AI model.

Steps for Completion

1. How can AI developers check the accuracy of AI models?
   a. ______________________________
   ____
2. What is robustness in relation to AI models?
   a. ______________________________
   ____
   ____
3. What is adversarial testing?
   a. ______________________________
   ____
   ____
   ____
4. What is edge case testing?
   a. ______________________________
   ____
5. How can AI developers see the true speed of an AI application?
   a. ______________________________
   ____
6. Determine whether each statement is true or false.
   a. ____________ For classification-based AI projects, AI developers should analyze false positives and false negatives.
   b. ____________ To best test the speed of an AI application, AI developers should establish speed benchmarks.
   c. ____________ Few AI applications are built with rapid development capabilities.
   d. ____________ AI applications should be compatible with multiple operating systems.
   e. ____________ AI developers should test AI applications using the fastest connections available.

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 4
Topic: Design a Production Pipeline
Subtopic: Test AI Accuracy; Test AI Robustness; Test AI Speed; Test App to Fit Size of Use Case

Objectives covered
5 Application Integrity and Deployment
5.1 Design a production pipeline, including application integration
5.1.4 Test accuracy, robustness, and latency of AI through application
5.1.5 Test application to fit size of use case (e.g. in AI for mobile applications)

Domain 5
Lesson 2

Artificial Intelligence

LearnKey

AI Model Challenges

AI models have many potential challenges. Understanding the types of challenges they are likely to encounter will help AI developers be better equipped to face those challenges when they happen, address the needs of those challenges, and ultimately succeed in building robust AI models. In addition to understanding the types of challenges AI developers face with AI models, they also need to understand the indicators of these challenges.

Purpose

Upon completing this project, you will better understand the potential challenges of implementing an AI model and how to mitigate those challenges.

Steps for Completion

1. List two types of challenges AI developers may encounter regarding data.
   a. ____
   ____
   ____
   ____
2. List two types of challenges AI developers may encounter regarding user acceptance.
   a. ____
   ____
   ____
3. What is data drift?
   a. ____
   ____
4. What is concept drift?
   a. ____
5. Determine whether each statement is true or false.
   a. ____________ Many organizations use on-premises servers for AI models because those infrastructures are more robust and scalable.
   b. ____________ If AI developers suspect bias in data, they should consider changing the data set to mitigate said bias.
   c. ____________ For overall performance issues, AI developers should consider revising the features included in an AI model.
   d. ____________ Feature adjustment should place emphasis on the variance in the feature itself, not whether the feature contributes well enough to an outcome.

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 5
Topic: Address Potential Challenges
Subtopic: Understand Types of Challenges; Understand Challenge Indicators; Understand Challenge Mitigations

Objectives covered
5 Application Integrity and Deployment
5.2 Identify potential challenges of models in productionand plan for AI solution support
5.2.1 Understand the types of challenges you are likely to encounter
5.2.2 Understand the indicators of challenges
5.2.3 Understand how each type of challenge could be mitigated

Documentation and Support

Documentation is vital in any IT project. As part of supporting an AI solution, teams must document it well to best allow for maintenance and understanding. Part of supporting an AI solution involves understanding the solution itself, its processes, and its goals. A knowledge base should be built using the best documents to support cases and similar happenings within the AI model.

Purpose

Upon completing this project, you will better understand the importance of AI documentation and support.

Steps for Completion

1. When building an AI model, what information should be documented?
    a. ______________________________
    b. ______________________________
    c. ______________________________
    d. ______________________________
    e. ______________________________
2. Determine whether each statement is true or false.
    a. __________ A trained support team must support an AI solution to ensure an AI model is used effectively.
    b. __________ In a small business, AI solution support often falls on the usual IT support team.
3. There must be good communication between a technical team and a support team. List at least two types of changes to an AI model that a technical team must report to its support team.
    a. ______________________________

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 4
Topic: Support the AI Solution
Subtopic: Document for Maintenance; Train a Support Team

Objectives covered
5 Application Integrity and Deployment
5.2 Identify potential challenges of models in productionand plan for AI solution support
5.2.4 Document the functions within the AI solution to allow for maintenance (updates, fixing bugs, handling edge cases)
5.2.5 Train a support team

Feedback Mechanisms

Another aspect of supporting an AI solution is requesting feedback on that AI solution from the people using it. There are many ways to implement feedback mechanisms. Support teams can choose the most effective mechanisms for their products and then use feedback to improve them.

Purpose

Upon completing this project, you will better understand the types of feedback mechanisms for AI solutions.

Steps for Completion

1. List at least two different mechanisms that can be built to generate feedback.
    a. ____
    ____
2. If a team does not have time to build a feedback mechanism, they should have a dedicated ______________ somebody can use to send feedback on an implemented AI system.
3. Teams may consider sending out ______________ or ______________ that people can fill out at specific time intervals after an AI solution has been implemented.
4. For large-scale projects, teams may want to conduct user ______________ and user ______________ both during and after implementation.
5. If the AI solution is public-facing, an organization may review the feedback on its ______________ channels to gauge whether the AI solution is working.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Support the AI Solution
Subtopic: Implement a Feedback Mechanism

Objectives covered
5 Application Integrity and Deployment
5.2 Identify potential challenges of models in productionand plan for AI solution support
5.2.6 Implement a feedback mechanism

LearnKey

Maintaining AI Solutions

Part of supporting an AI solution can include implementing a drift detector or developing other ways to gather data. A drift detector monitors and detects changes in data distribution or model performance and identifies shifts in the underlying data used in an AI solution. Drift detection aims to ensure an AI model stays aligned with the evolving dynamics of the data being used within the model. It enhances the accuracy and precision of an AI model. Implementing new data collection methods aims to make an AI model more accurate in its overall outcomes for an AI-based problem.

Purpose

Upon completing this project, you will better understand drift detection and other ways to maintain AI solutions.

Steps for Completion

1. Detecting drift involves monitoring changes or variations in the patterns and distribution of the ______________________________ categories used.
2. When a team implements a drift detector, they can define thresholds for acceptable changes in data and generate ______________________ when performance falls outside acceptable ranges.
3. What data types should be gathered to update and retrain an AI model?
   a. ____
      ____
      ____
      ____
4. For some AI models, a ____________________ may need to be created or adjusted to better reflect a model's needs.
5. The structure and nature of ______________________ data collection may require a different computational setup to ensure seamless integration with an AI model versus historical data.
6. When implementing new data collection methods, one should ensure that data is collected in compliance with data _________________________ and security regulations.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 4
Topic: Support the AI Solution
Subtopic: Implement Drift Detector; Implement Ways to Gather New Data

Objectives covered
5 Application Integrity and Deployment
5.2 Identify potential challenges of models in productionand plan for AI solution support
5.2.7 Implement drift detector
5.2.8 Implement ways to gather new data

Domain 5
Lesson 3

Artificial Intelligence

LearnKey

Access Levels and Permissions

Creating an effective security plan is an integral part of an AI solution. The security plan should include a thorough analysis of potential risks an AI system could encounter, along with internal access levels or permissions on data. An organization needs to identify who requires access to data and ensure that only authorized personnel can access it.

Purpose

Upon completing this project, you will become more familiar with access levels and permissions.

Steps for Completion

1. Where can one find the permissions granted to specific groups on a Windows file folder?
    a. ____
    ____
    ____
2. How is Role-Based Access Control (RBAC) easier to manage than assigning permissions to individuals or groups?
    a. ____
    ____
    ____
3. Why is it important to store data used for AI models in a secure location?
    a. ____
    ____
    ____
4. Determine whether the statement is true or false.
    a. ____________ Ensuring data is secure and only accessed by those who need access to it minimizes possible data breaches.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Build a Security Plan
Subtopic: Access Levels and Permissions

Objectives covered
5 Application Integrity and Deployment
5.3 Build a security plan
5.3.1 Consider internal access levels or permissions

Infrastructure Security and Risk Models

Infrastructure security encompasses both network and physical security, such as data and the network on which it is stored. It is important to protect infrastructure against both physical and cyber threats. When real-time learning models are used, a specific type of security consideration is required because data in transit is subject to attack.

Purpose

Upon completing this project, you will better understand infrastructure security.

Steps for Completion

1. What are two ways a network can be set up to help protect data from being compromised via intrusion or attack?
   a. ______________________________
   ____
2. Determine whether each statement is true or false.
   a. __________ Data only needs to be encrypted in transit to protect it from being stolen.
   b. __________ Only those with access to data should have access to the server containing the data.
3. What type of data do real-time learning models collect and send?
   a. ______________________________
   ____
4. What are two ways to identify risks for attack?
   a. ______________________________
   ____
5. What is threat modeling?
   a. ______________________________
   ____
6. What are risk assessments?
   a. ______________________________
   ____
7. What could a spike in data indicate in an AI model?
   a. ______________________________
   ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 5
Topic: Build a Security Plan
Subtopic: Infrastructure Security; Assess Risk Models

Objectives covered
5 Application Integrity and Deployment
5.3 Build a security plan
5.3.2 Consider infrastructure security
5.3.3 Assess the risk of using a certain model or potential attack surfaces (e.g., adversarial attacks on real-time learning model)

Domain 5
Lesson 4

Artificial Intelligence

LearnKey

Train Users

Deploying an application with an AI model includes training those who are going to use the model, what to expect from it, and how to use the application. In addition to model limitations, people using an AI model should know the intended usage of a model. Documentation is an important step that helps prepare users to use an application. For an AI model to be successful, AI developers must manage customer expectations as well.

Purpose

Upon completing this project, you will better understand how to prepare users to use an application with an AI model.

Steps for Completion

1. What should be available to all users of an application used to generate AI results?
   a. ____
   ____
   ____

2. Why should AI developers communicate the limitations of an AI model to users and stakeholders?
   a. ____
   ____
   ____

3. Why should AI developers communicate the intended use of an AI model to users and stakeholders?
   a. ____
   ____
   ____

4. List three items that can be included in documentation.
   a. ____
   ____

5. Where should documentation that is suitable for the public be stored?
   a. ____
   ____

6. What two locations are commonly used to store documentation that is not suitable for the public?
   a. ____
   ____

7. What are two ways in which AI developers can manage customer expectations?
   a. ____
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
15 minutes

Video reference
Domain 5
Topic: Train Customers on Product
Subtopic: Inform on Model Limitations; Inform of Intended Model Usage; Share Documentation; Manage Customer Expectations

Objectives covered
5 Application Integrity and Deployment
5.4 Train customers on how to use product and what to expect from it
5.4.1 Inform users of model limitations
5.4.2 Inform users of intended model usage
5.4.3 Share documentation
5.4.4 Manage customer expectations

Domain 6
Lesson 1

Artificial Intelligence

Monitor Performance

To monitor the performance of an AI app or model, the first step is to implement logging and monitoring. Logging performance facilitates multiple important areas, such as security and accountability. A variety of robust monitoring systems should be used to ensure high performance, and administrators should set up alerts to notify them of any issues.

Purpose

Upon completing this project, you will better understand the purpose of logging, monitoring systems, and alerts in maintaining AI performance.

Steps for Completion

1. What are the four main areas of logging?
   a. ____
   ____
2. Why is it important to log performance?
   a. ____
   ____
   ____
3. List three types of monitoring systems that should be used to oversee an AI model.
   a. ____
   ____
4. Determine whether each statement is true or false.
   a. ____________ Alerts should be well-documented, along with any troubleshooting steps taken to address them.
   b. ____________ Alerts should be addressed in batches once a significant number of them have occurred.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Engage in Oversight
Subtopic: Log App and Model Performance; Use Robust Monitoring Systems; Act Upon Alerts

Objectives covered
6 Maintaining and Monitoring AI in Production
6.1 Engage in oversight
6.1.1 Log and monitor application and model performance to facilitate security, debug, accountability, and audit
6.1.2 Use robust monitoring systems
6.1.3 Act upon alerts

Drift and Degraded Modes

As part of engaging in oversight, an AI system must be observed over time and in various contexts. The purpose of this is to see if the system is operating with any drift or degraded modes of operation. These issues need to be prevented to ensure an AI model runs smoothly and produces accurate results.

Purpose

Upon completing this project, you will better understand how to detect and resolve drift and degraded modes in an AI system.

Steps for Completion

1. What is drift?
   a. ______________________________
   ____
2. How should drift be addressed?
   a. ______________________________
   ____
3. What is a degraded mode of operation?
   a. ______________________________
   ____
4. How should a degraded mode of operation be addressed?
   a. ______________________________
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Engage in Oversight
Subtopic: Drift and Degraded Mode

Objectives covered
6 Maintaining and Monitoring AI in Production
6.1 Engage in oversight
6.1.3 Observe the system over time in a variety of contexts to check for drift or degraded modes of operation

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Failure Detection

While overseeing an AI system, administrators need to watch for any issues with the system failing to support new information. This issue is often caused by information not importing into the system properly, or other issues with data coming into the system and being transformed.

Purpose

Upon completing this project, you will better understand how to troubleshoot an AI system that fails to support new information.

Steps for Completion

1. What are two indicators that new information has failed to make its way into an AI system?
    a. ______________________________
2. How should administrators troubleshoot issues with external data sources?
    a. ______________________________
3. How should administrators troubleshoot issues with user interactions?
    a. ______________________________

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Engage in Oversight
Subtopic: Detect Failure to Support New Info

Objectives covered
6 Maintaining and Monitoring AI in Production
6.1 Engage in oversight
6.1.4 Detect any way system fails to support new information

Evaluate Outputs and Document Results

Evaluating outputs can help confirm that an AI model and its data adhere to regulatory requirements, if applicable. However, not all industries need to worry about regulatory compliance when working with data. Part of adhering to regulatory requirements is keeping solid documentation of any adjustments made to AI models.

Purpose

Upon completing this project, you will better understand evaluating outputs.

Steps for Completion

1. What is the GDPR?
    a. ____
    ____
2. What is the first step in evaluating outputs to comply with data requirements' thresholds?
    a. ____
3. What should one do if they find issues of non-compliance with regulations?
    a. ____
    ____
    ____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 6
Topic: Confirm Regulatory Requirements
Subtopic: Evaluate Outputs; Document Results

Objectives covered
6 Maintaining and Monitoring AI in Production
6.1 Engage in oversight
6.1.5 Evaluate outputs according to thresholds defined in regulatory requirements

Domain 6
Lesson 2

Artificial Intelligence

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Assessing Business Impact

To assess the business impact of an AI system, administrators need to establish key performance indicators (KPIs). Once KPIs are established, they should be used to compare the results of changes that are made to determine if any adjustments are needed. Administrators should also be on the lookout for unexpected metrics and make sure to troubleshoot them appropriately.

Purpose

Upon completing this project, you will understand how to use KPIs to assess the business impact of an AI system.

Steps for Completion

1. To build key performance indicators, a ________________ measurement needs to be established first.
2. Why is it important to establish and monitor KPIs?
    a. ____
    ____
    ____
    ____
    ____
3. Why should new metrics be compared to previous metrics when changes are made?
    a. ____
    ____
    ____
4. How should unexpected metrics be addressed?
    a. ____
    ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Assess Business Impact
Subtopic: Track Impact Metrics; Compare Previous to New Metrics; Act on Unexpected Metrics

Objectives covered
6 Maintaining and Monitoring AI in Production
6.2 Assess business impact (key performance indicators)
6.2.1 Track impact metrics to determine whether the solution has solved the problem
6.2.2 Compare previous metrics with new metrics when changes are made
6.2.3 Act on unexpected metrics by finding problem and fixing it

Domain 6
Lesson 3

Artificial Intelligence

Individual and Community Impact

In addition to assessing business impacts, it is important to evaluate an AI system's impacts on individuals or communities. It is particularly important to ensure subgroups are not affected by bias. It is also useful to evaluate the system for optimization to ensure that it is as functional as possible.

Purpose

Upon completing this project, you will better understand how to assess and address the individual and community impacts of an AI system.

Steps for Completion

1. Why is it important to analyze data collected regarding impact on various subgroups?
   a. ______________________________
   ____
2. What are two methods of identifying potential issues for individuals and communities?
   a. ______________________________
   ____
   ____
   ____
3. How should disparities in outcomes be addressed once they are found?
   a. ______________________________
   ____
4. What are two methods of identifying opportunities for observation?
   a. ______________________________
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Measure Impacts
Subtopic: Analyze Subgroup Impact; Identify and Mitigate Issues; Identify Optimization Opportunities

Objectives covered
6 Maintaining and Monitoring AI in Production
6.3 Measure impacts on individuals and communities
6.3.1 Identify impact on specific subgroups
6.3.2 Identify and mitigate issues
6.3.3 Identify opportunities for optimization

Domain 6
Lesson 4

Artificial Intelligence

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User Feedback

User feedback is a crucial part of shaping and adjusting an AI model. Various methods can help assess user satisfaction and user confusion, including both directly asking users and observing activity within an app. Once feedback has been received, it can be used to create, test, and implement changes that will improve users' experiences with the AI model.

Purpose

Upon completing this project, you will better understand how to collect and use user feedback to improve an AI model.

Steps for Completion

1. What is a Net Promoter Score (NPS)?
   a. ______________________________
   ____
   ____
2. How can usage metrics help measure user satisfaction?
   a. ______________________________
   ____
   ____
3. What is A/B testing?
   a. ______________________________
   ____
   ____
4. What are three methods of assessing user confusion?
   a. ______________________________
   ____
5. What steps should be taken to incorporate user feedback once it is collected?
   a. ______________________________
   ____
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Handle User Feedback
Subtopic: Measure User Satisfaction; Assess Whether Users Are Confused; Incorporate Feeback into Future Versions

Objectives covered
6 Maintaining and Monitoring AI in Production
6.4 Handle feedback from users
6.4.1 Measure user satisfaction
6.4.2 Assess whether users are confused (e.g., do they understand what the AI is supposed to do for them?)
6.4.3 Incorporate feedback into future versions

Domain 6
Lesson 5

Artificial Intelligence

LK LearnKey

Improve or Decommission

Once an AI is implemented, it should be regularly evaluated to determine whether it needs to be improved or decommissioned. Impact observations regarding business, community, and technological impacts help with making these decisions. An AI that works well can continue to be used as is, while an AI that no longer performs as it should needs to be retrained or decommissioned, depending on the issue.

Purpose

Upon completing this project, you will better understand how to use impact observations to decide whether to improve or decommission an AI.

Steps for Completion

1. What are two business impacts to assess for an AI model?
   a. ______________________________
   ____
2. What are two community impacts to assess for an AI model?
   a. ______________________________
   ____
3. What technology trends should be assessed for an AI model?
   a. ______________________________
   ____
   ____
4. When should an AI model be retrained?
   a. ______________________________
5. When should an AI model be discontinued?
   a. ______________________________
   ____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 6
Topic: Improve or Decommission
Subtopic: Combine Impact Observations; Retrain, Continue, or Decommission

Objectives covered
6 Maintaining and Monitoring AI in Production
6.5 Evaluate operation on a regular basis
6.5.1 Combine impact observations (e.g., business, community, technology trends) to assess AI value
6.5.2 Decide whether to retrain AI, continue to use AI as is, or to decommission AI

Appendix

Artificial Intelligence

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Glossary

Domain 1

Term	Definition
Accuracy	The percentage of correct predictions within a model.
AI	AI (artificial intelligence) is intelligence that can be programmed to mimic human intelligence.
Algorithm	A set of rules used to generate calculations or perform problem-solving operations.
Attack Surface	The parts of data or a system that are vulnerable to attack.
AUC	AUC (Area Under the Curve) is a model's probability of choosing a positive instance over a negative instance.
Bias	An imbalance in data that can cause data to be skewed toward a demographic group, which can harm an AI machine-learning model.
Classification	A model type in which a result is categorized, usually with one of two values, such as a yes or no value.
Data at Rest	Data stored on an on-premises or cloud storage drive.
Data in Transit	Data in the process of being copied or moved from one location to another.
Explainability Methods	Methods used to help non-technical people understand an AI solution's aspects.
F1 Score	The harmonic mean of precision and recall.
Feasibility Analysis	In terms of AI solutions, an anaylsis that helps determine the likelihood of an AI project being accomplished
Feature	An input to an AI model used to generate output.
Labeled Data	Data that is paired with output targets in an AI model.
Linear Regression	A form of regression in which an outcome is a continuous variable, like a number.
Logistic Regression	A form of regression in which an outcome uses binary dependent variables.
Machine Learning	A process by which a system trains an AI model.
Model	An app used to generate predictions and outcomes based on the entering and training of data entered into the app.
Natural Language Processing	A form of AI that reads data from text and images and can include speech recognition and object detection.
Precision	The percentage of true positive predictions among all positive predictions in an outcome.
Pre-Trained Model	An AI model that has been trained for general responses to data requests and generative data requests.
Power Virtual Agents	A Microsoft tool used to build chatbots.
Recall	The measure of true positive predictions among all actual positives.
Regression	A model type that looks at the relationship between a dependent variable and one or more independent variables.
Reinforcement Learning	A type of machine learning model that involves an AI agent performing actions that maximize a certain reward.
Risk Register	A document that contains known risks for a project and the impact, probability, and mitigation strategy for those risks.
ROC	ROC (Receiver Operating Characteristic Curve) is the plot of the True Positive Rate against the False Positive Rate for model results.

Term	Definition
R-Squared	A value that measures the proportion of variance of a dependent variable that is predictable from independent variables.
Scalability	The ability to adjust machine resources up or down depending upon workload demand for one or more apps on that machine.
SME	SME (Subject Matter Expert) is an expert on the problem or problems attempting to be solved through the use of an AI model.
Stakeholder	An individual, group, or organization that may affect or be affected by a project.
Transparency	In the context of AI, the act of being forthright about how data is collected and used for AI models.
Unlabeled Data	Data that is not paired with output targets in an AI model.
Unsupervised	A type of machine learning model that looks for hidden patterns or structures in data.

Domain 2

Term	Definition
Annotation	The labeling of data so that it can train an AI model properly.
Anomaly	An unusual activity in a group of otherwise normal activities.
Azure Machine Learning	A cloud-based tool used to build AI models.
Binary	A form of data that stores data as ones and zeros.
Data Type	A characteristic of data, such as numeric, string, or date.
Feature Vector	An ordered list of numerical properties of observed phenomena.
One-Hot Encoding	A form of encoding that assigns categorical variables to binary numbers.
OpenRefine	A tool used to cleanse data to ready it for an AI learning model.
Sentiment Analysis	A form of machine learning in which data is read and assigned a value for its positivity or negativity.
Tokenization	The process of converting words to numbers to be used for data in an AI model.
Trifacta Wrangler	A tool used to cleanse data to ready it for an AI learning model.

Domain 3

Term	Definition
Assumption	A belief or condition taken to be true for a system to work as intended.
Constraint	A limitation or restriction on an AI system.
Correlation	A measure of connection between two points of data.
Derived Feature	A feature built using a formula, through splitting or combining data, or through assigning keywords to data based on conditions.
Feature Vector	An ordered list of numerical properties of observed phenomena.
GDPR	GDPR (General Data Protection Regulation) is the European Union (EU) standard for handling data that affects companies doing business in the EU.
Maintenance Logs	A form of documentation that shows how tests for an AI model are built and executed
Predicate	A logical statement or condition defining properties or relationships between different entities in an AI system.
Project Scope	A project management document that defines the work to be done on a project, how it will be done, and the work that will not be included in the project.
Split Ratio	The ratio of data being used for training an AI model versus the data used for testing the model.
Transformation	In the context of data, the process of cleaning data, converting formats, normalizing numbers, making data binary for yes/no situations, and data aggregation.
Test Documentation	A form of documentation that shows how tests for an AI model are built and executed

Domain 4

Term	Definition
Column Chart	A chart that compares values across multiple categories and series' of data.
Confusion Matrix	A chart that defines true and false positives and true and false negatives for an AI model.
Decision Tree Algorithm	A machine learning method in which samples are split into two or more sets of data based on input variable differentiators.
Explainability Requirement	A clear explanation for a decision, prediction, or recommendation made from an AI model.
False Negative	An incorrect negative result for an AI model.
False Positive	An incorrect positive result for an AI model.
Histogram	A chart that represents the distribution of numerical data.
Imbalanced Data	A dataset that has too much data from one or more groups of data.
Interpretability	The ability for one to understand the results of an AI model.
Iteration	A sequence of training an AI model, often needs to be repeated to fully train an AI model.
K-Means Clustering	A machine learning model that splits data into an unspecified number of groups.
KPI	KPI (Key Performance Indicator) is a visualization that sets standards for performance for, in the context of AI, AI models.
Line Chart	A chart full of lines that often shows time-based data comparisons.
Neural Network	A type of algorithm designed to mimic the human brain.
Overfitting	A situation in which a model learns training data very well but performs poorly on new, unseen data.
Parameter	An input value used to help train an algorithm.

Term	Definition
Performance Metrics	Data gathered on an AI model and used to ensure that an AI model is working properly.
Pie Chart	A chart that shows how, for a single category, values compare to each other and as a whole.
Self-Fulfilling Prophecy	A phenomenon where biased data perpetuates a wanted or perceived bias within an AI model.
Sensitivity	The relative change of a model and its output.
Specificity	The process of evaluating a model's ability to accurately identify negative instances or nonrelevant outcomes.
Stakeholder	An individual, group, or organization that may affect or be affected by a project.
Supervised Algorithm	An algorithm that uses labeled data, which means data where a target variable is known.
Trend	A pattern of data within an AI model.
True Negative	A correct negative result for an AI model.
True Positive	A correct positive result for an AI model.
Underfitting	A situation in which a model does not perform well on training data or new, unseen data.
Unsupervised Algorithm	An algorithm that does not use labeled data.
Visualization	A chart or report that explains the results of an AI model or the data used to build an AI model.

Domain 5

Term	Definition
Adversarial Testing	A test in which data is intentionally manipulated to evaluate an AI system's vulnerability to adversarial attacks or misleading inputs.
API	A set of rules that allows applications to communicate with each other.
Attack Surface	The parts of data or a system that are vulnerable to attack.
CD	Continuous Deployment (CD) is a workflow in which code is released manually or automatically once it passes checks.
CI	Continuous Integration (CI) is a workflow in which code developers regularly merge new code in existing code.
Concept Drift	The change in the underlying concepts used to train AI models.
Data Drift	The change in the statistical properties of the input data within an AI model.
Drift Detector	A mechanism that identifies shifts in the underlying data used in an AI solution.
Edge Case Testing	A test in which an AI system's performance is pushed beyond its normal operating range.
Feedback Mechanism	A means by which feedback is collected for an AI system, whether that be through forms, a website, or directly within the system.
GitHub	A repository used to store code for development projects.
JSON	JSON (JavaScript Object Notation) is a file format in which data is stored in key-value pairs.
Latency	The time it takes for a request for data to be fulfilled.
Pipeline	A series of steps used to transform data used for an AI model into a final prediction or outcome.
RBAC	RBAC (Role-Based Access Control) is a permission standard in which permissions are assigned to roles, and roles are assigned to individuals and groups.
SQL Server	Microsoft's database server and often used to store data used to generate a pipeline for an AI model.
User Acceptance	The want of people to use an AI model and the outcomes it produces.

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Domain 6

Term	Definition
A/B Test	A test in which different app versions are shown to different people and usage patterns and responses are compared to each other.
Decommission	The act of no longer using an AI model in production due to its lack of effectiveness.
Degraded Mode of Operation	A mode of operation in which an AI model is not performing to its maximum capabilities.
Log	A list of activities in an AI model that can be used to identify potential security problems with a model.
Usage Metrics	The extent to which people use an AI model to its fullest.

Objectives

Artificial Intelligence Objectives		
Domain 1 AI Problem Definition	**Domain 2 Data Collection and Transformation**	**Domain 3 Feature Engineering**
1.1 Identify the problem you are trying to solve using AI (e.g., user segmentation, improving customer service) 1.1.1 Identify the user persona – who is the solution for? 1.1.2 Identify the need that will be addressed by the solution 1.1.3 Consider input and output requirements and constraints of the solution 1.1.4 Determine whether AI is called for 1.1.5 Identify whether interaction between AI and humans is required 1.1.6 Consider advantages and disadvantages of using AI in the situation 1.1.7 Define measurable success 1.1.8 Benchmark against domain or organization-specific risks to which the project may be susceptible	2.1 Identify data sources and requirements 2.1.1 Determine type/characteristics of data needed 2.1.2 Decide if there is an existing dataset or if you need to generate your own 2.1.3 Choose data collection methods 2.1.4 When generating your own dataset, decide whether collection can be automated or requires user input"	3.1 Select features for the AI model 3.1.1 Determine which data features to include 3.1.2 Build initial feature vectors for test/train dataset 3.1.3 Consult with subject-matter experts to confirm feature selection
1.2 Identify the areas of expertise and resource requirements needed to solve the problem 1.2.1 Identify business expertise required 1.2.2 Identify the need for domain (subject-matter) expertise on the problem 1.2.3 Identify AI expertise needed 1.2.4 Identify implementation expertise needed 1.2.5 Assess whether the problem is solvable with available computing resources 1.2.6 Consider the budget of the project and the resources that are available	2.2 Assess data quality 2.2.1 Identify missing or corrupt data elements 2.2.2 Examine collection techniques for potential sources of bias 2.2.3 Make sure the amount and variety of data is enough to build an unbiased model 2.2.4 Assess the quality of the annotation	3.2 Engage in feature engineering 3.2.1 Review features and determine what standard transformations are needed 3.2.2 Create processed datasets 3.2.3 Identify whether transformations have been applied correctly
1.3 Classify the problem (e.g., regression, unsupervised learning) 1.3.1 Examine available data 1.3.2 Determine problem type (e.g., classification, regression, unsupervised, reinforcement) 1.3.3 Recognize when a pre-trained model is appropriate, if you can fine-tune an existing model, or if you need to train the model from scratch 1.3.4 Conduct feasibility analysis to determine whether it can be realistically modeled	2.3 Convert data into suitable formats (e.g., numerical, image, time series) 2.3.1 Convert data to binary (e.g., images become pixels) 2.3.2 Convert computer data into features suitable for AI (e.g., sentences become tokens) 2.3.3 Annotate the data if necessary	3.3 Identify training and test datasets 3.3.1 Separate available data into training and test datasets 3.3.2 Ensure test dataset is representative
1.4 Plan for AI to be used responsibly 1.4.1 Identify potential ways that the AI can mispredict and harm specific user groups 1.4.2 Set guidelines for data gathering and use 1.4.3 Set guidelines for algorithm selection based on user requirements 1.4.4 Consider how the user of the solution can interpret the results		3.4 Document data decisions 3.4.1 List assumptions, predicates, and constraints upon which design choices have been reasoned 3.4.2 Make this information available to regulators and end users who demand deep transparency

Artificial Intelligence Objectives		
Domain 1 AI Problem Definition	Domain 2 Data Collection and Transformation	Domain 3 Feature Engineering
1.4.5 Consider out-of-context use of AI results		3.4.3 Identify situations in which particular types of documentation are relevant
1.5 Choose transparency and validation activities 1.5.1 Communicate the intended purpose of data collection 1.5.2 Decide who should see the results 1.5.3 Review legal requirements specific to the industry with the problem being solved 1.5.4 Describe explainability methods for communicating AI decisions to stakeholders		

Artificial Intelligence Objectives		
Domain 4 **AI Algorithms and Models**	**Domain 5** **Application Integrity and Deployment**	**Domain 6** **Maintaining and Monitoring AI in Production**
4.1 Consider applicability of specific algorithms 4.1.1 Evaluate AI algorithm families: 4.1.1.1 Supervised learning 4.1.1.2 Unsupervised learning 4.1.1.3 Reinforcement learning 4.1.1.4 Natural language processing (NLP) 4.1.1.5 Regression 4.1.2 Decide which algorithms are suitable, e.g., neural network, decision tree, k means	5.1 Design a production pipeline, including application integration 5.1.1 Plan for continuous integration and deployment (CI/CD) 5.1.2 Design a pipeline that integrates with existing data stores and connects to the application 5.1.3 Build the connection between the AI and the application 5.1.4 Test accuracy, robustness, and latency of AI through application 5.1.5 Test application to fit the size of use case (e.g., in AI for mobile applications)	6.1 Engage in oversight 6.1.1 Log and monitor application and model performance to facilitate security, debug, accountability, and audit 6.1.2 Act upon alerts 6.1.3 Observe the system over time in a variety of contexts to check for drift or degraded modes of operation 6.1.4 Detect ways the system fails to support new information 6.1.5 Evaluate outputs according to thresholds defined in regulatory requirements
4.2 Train a model using the selected algorithms 4.2.1 Train model using an algorithm with starting parameters 4.2.2 Tune the model by changing parameters 4.2.3 Identify performance metrics for the model 4.2.4 Determine whether additional iterations are required	5.2 Identify potential challenges of models in production and plan for AI solution support 5.2.1 Understand the types of challenges you are likely to encounter 5.2.2 Understand the indicators of challenges 5.2.3 Understand how each type of challenge could be mitigated 5.2.4 Document the functions within the AI solution to allow for maintenance (updates, fixing bugs, handling edge cases) 5.2.5 Train a support team 5.2.6 Implement a feedback mechanism 5.2.7 Implement drift detector 5.2.8 Implement ways to gather new data	6.2 Assess business impact (key performance indicators) 6.2.1 Track impact metrics to determine whether the solution has solved the problem 6.2.2 Compare previous metrics with new metrics when changes are made 6.2.3 Act on unexpected metrics by finding problems and fixing them
4.3 Evaluate model accuracy, precision, and sensitivity 4.3.1 Check for overfitting and underfitting 4.3.2 Generate metrics or KPIs (accuracy, precision, recall, F1 score, etc.) 4.3.3 Introduce new test data to cross-validate robustness, testing how model handles unforeseen data 4.3.4 Interpret performance metrics (sensitivity, specificity, confusion matrix) 4.3.5 Identify appropriate techniques to improve model performance	5.3 Build a security plan 5.3.1 Consider internal access levels or permissions 5.3.2 Consider infrastructure security 5.3.3 Assess the risk of using a certain model or potential attack surfaces (e.g., adversarial attacks on real-time learning models)	6.3 Measure impacts on individuals and communities 6.3.1 Identify impact on specific subgroups 6.3.2 Identify and mitigate issues 6.3.3 Identify opportunities for optimization
4.4 Identify potential sources of bias 4.4.1 Verify that inputs resemble training data 4.4.2 Confirm that training data do not contain irrelevant correlations we do not want the classifier to rely on 4.4.3 Check for imbalances in data 4.4.4 Guard against creating self-fulfilling prophecies based upon historical biases	5.4 Train customers on how to use product and what to expect from it 5.4.1 Inform users of model limitations 5.4.2 Inform users of intended model usage 5.4.3 Share documentation 5.4.4 Manage customer expectations	6.4 Handle feedback from users 6.4.1 Measure user satisfaction 6.4.2 Assess whether users are confused (e.g., do they understand what the AI is supposed to do for them?) 6.4.3 Incorporate feedback into future versions

Artificial Intelligence Objectives		
Domain 4 **AI Algorithms and Models**	**Domain 5** **Application Integrity and Deployment**	**Domain 6** **Maintaining and Monitoring AI in Production**
4.5 Select specific model after experimentation 4.5.1 Consider cost, speed, and other factors when evaluating models 4.5.2 Determine whether the selected model meets explainability requirements		6.5 Evaluate operation on a regular basis 6.5.1 Combine impact observations (e.g., business, community, technology trends) to assess AI value 6.5.2 Decide whether to retrain AI, continue to use AI as-is, or to decommission AI
4.6 Tell data stories and obtain stakeholder approval 4.6.1 Identify which visualization types will represent the data story most appropriately 4.6.2 Where feasible, create visualizations of the results 4.6.3 Look for trends 4.6.4 Verify that visualizations are useful for making a decision 4.6.5 Collect results and benchmark risks and hold sessions to evaluate the solution		

Lesson Plan

Approximately 23.5 hours of video, labs, and projects

Artificial Intelligence

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Domain 1 Lesson Plan

Domain 1 - AI Problem Definition [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	AI Problem Definition: Pre-Assessment			
Lesson 1 Video time - 00:12:46 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Identify the Problem** How to Study for the Exam Identify the User Persona Identify the Need to Be Addressed Find Out Input and Output Determine if AI Is Needed	1.1 Identify the problem you are trying to solve using AI (e.g., user segmentation, improving customer service) 1.1.1 Identify the user persona – who is the solution for? 1.1.2 Identify the need that will be addressed by the solution 1.1.3 Consider input and output requirements and constraints of the solution 1.1.4 Determine whether AI is called for	N/A	Identify User Persona, Need, Input, and Output – pg. 9 N/A Determine if AI is Needed – pg. 10 N/A
Lesson 2 Video time - 00:08:49 Exercise Lab time - 00:00:00 Workbook time - 00:30:00	**Decisions and Benchmarks** See if Human Interaction is Needed Consider Upsides and Downsides Define Measurable Success Benchmark Against Risks	1.1 Identify the problem you are trying to solve using AI (e.g., user segmentation, improving customer service) 1.1.5 Identify whether interaction between AI and humans is required 1.1.6 Consider advantages and disadvantages of using AI in the situation 1.1.7 Define measurable success 1.1.8 Benchmark against domain or organization-specific risks to which the project may be susceptible	N/A	Human Interaction – pg. 12 N/A Upsides and Downsides of AI – pg. 13 N/A Define Measurable Success – pg. 14 N/A Benchmark Against Risk – pg. 15 N/A
Lesson 3 Video time - 00:06:27 Exercise Lab time - 00:00:00 Workbook time - 00:10:00	**Identify Needed Expertise** Identify Business Expertise Identify Domain Expertise Identify AI Expertise Identify Implementation Expertise	1.2 Identify the areas of expertise and resource requirements needed to solve the problem 1.2.1 Identify business expertise required 1.2.2 Identify the need for domain (subject-matter) expertise on the problem 1.2.3 Identify AI expertise needed 1.2.4 Identify implementation expertise needed	N/A	Identify Business and Domain Expertise – pg. 17 N/A Identify AI and Implementation Expertise – pg. 18 N/A
Lesson 4 Video time - 00:06:58 Exercise Lab time - 00:00:00 Workbook time - 00:10:00	**Identify Resource Requirements** Assess Existing Resources Consider Budget and Resources	1.2 Identify the areas of expertise and resource requirements needed to solve the problem 1.2.5 Assess whether the problem is solvable with available computing resources 1.2.6 Consider the budget of the project and the resources that are available	N/A	Buy or Develop – pg. 20 N/A
Lesson 5 Video time - 00:09:40 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Classify the Problem** Examine Available Data Determine Problem Type Examine Model Needs Conduct Feasibility Needs	1.3 Classify the problem (e.g., regression, unsupervised learning) 1.3.1 Examine available data 1.3.2 Determine problem type (e.g., classification, regression, unsupervised, reinforcement) 1.3.3 Recognize when a pre-trained model is appropriate, if you can fine-tune an existing model, or if you need to train the model from scratch 1.3.4 Conduct feasibility analysis to determine whether it can be realistically modeled	N/A	Classify the Problem – pg. 22 N/A Model and Feasibility Needs – pg. 23 N/A

Domain 1 - AI Problem Definition [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Lesson 6 Video time - 00:12:54 Exercise Lab time - 00:00:00 Workbook time - 00:30:00	**Ensure Appropriate Use of AI** Identify Potential User Group Harm Set Data Guidelines Set Algorithm Guidelines Consider Data Interpretation Consider Out-of-Context Use	1.4 Plan for AI to be used responsibly 1.4.1 Identify potential ways that the AI can mispredict and harm specific user groups 1.4.2 Set guidelines for data gathering and use 1.4.3 Set guidelines for algorithm selection based on user requirements 1.4.4 Consider how the user of the solution can interpret the results 1.4.5 Consider out-of-context use of AI results	N/A	Identify Potential User Group Harm – pg. 25 N/A Set Data Guidelines – pg. 26 N/A Set Algorithm Guidelines – pg. 27 N/A Data Interpretation – pg. 28 N/A Out-of-Context Use – pg. 29 N/A
Lesson 7 Video time - 00:07:34 Exercise Lab time - 00:00:00 Workbook time - 00:20:00	**Transparency and Validation** Communicate Intended Purpose Decide Who Should See Results Review Legal Requirements Describe Explainability Methods	1.5 Choose transparency and validation activities 1.5.1 Communicate the intended purpose of data collection 1.5.2 Decide who should see the results 1.5.3 Review legal requirements specific to the industry with the problem being solved 1.5.4 Describe explainability methods for communicating AI decisions to stakeholders	N/A	Transparency in Data Collection and Results – pg. 31 N/A Review Legal Requirements – pg. 32 N/A Describe Explainability Methods – pg. 33 N/A
Post-Assessment Assessment time - 01:00:00	AI Problem Definition: Post-Assessment			

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Domain 2 Lesson Plan

Domain 2 - Data Collection and Transformation [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Data Collection and Transformation: Pre-Assessment			
Lesson 1 Video time - 00:07:20 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Choose the Way to Collect Data** Determine Data Types Needed Decide if Needed Data Exists Choose Data Collection Methods Automated vs. User Input	2.1 Identify data sources and requirements 2.1.1 Determine type/characteristics of data needed 2.1.2 Decide if there is an existing dataset or if you need to generate your own 2.1.3 Choose data collection methods 2.1.4 When generating your own dataset, decide whether collection can be automated or requires user input	N/A	Data Collection – pg. 35 N/A
Lesson 2 Video time - 00:10:38 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Assess Data Quality** Look for Missing or Corrupt Data Examine Collection Techniques Ensure Enough Data Is Unbiased Assess the Quality of the Annotation	2.2 Assess data quality 2.2.1 Identify missing or corrupt data elements 2.2.2 Examine collection techniques for potential sources of bias 2.2.3 Make sure the amount and variety of data is enough to build an unbiased model 2.2.4 Assess the quality of the annotation	N/A	Clean Data – pg. 37 N/A Bias in Data – pg. 38 N/A
Lesson 3 Video time - 00:07:40 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Convert Data** Convert to Binary Convert to Features Suitable for AI Annotate the Data if Necessary	2.3 Convert data into suitable formats (e.g., numerical, image, time series) 2.3.1 Convert data to binary (e.g., images become pixels) 2.3.2 Convert computer data into features suitable for AI (e.g., sentences become tokens) 2.3.3 Annotate the data if necessary	N/A	Convert Images to Binary Format – pg. 40 N/A Convert Sentences to Tokens – pg. 41 N/A Data Annotation – pg. 42 N/A
Post-Assessment Assessment time - 01:00:00	Data Collection and Transformation: Post-Assessment			

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Domain 3 Lesson Plan

Domain 3 - Feature Engineering [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Feature Engineering: Pre-Assessment			
Lesson 1 Video time - 00:11:21 Exercise Lab time - 00:00:00 Workbook time - 00:20:00	**Select Features for the AI Model** Determine Features to Include Build Initial Feature Vectors Build Datasets Consult with SMEs on Selection	3.1 Select features for the AI model 3.1.1 Determine which data features to include 3.1.2 Build initial feature vectors for test/train dataset 3.1.3 Consult with subject-matter experts to confirm feature selection	N/A	Features – pg. 44 N/A Feature Vectors – pg. 45 N/A
Lesson 2 Video time - 00:05:33 Exercise Lab time - 00:04:00 Workbook time - 00:10:00	**Engage in Feature Engineering** Determine Needed Transformations Create Processed Datasets Identify Transformation Application	3.2 Engage in feature engineering 3.2.1 Review features and determine what standard transformations are needed 3.2.2 Create processed datasets 3.2.3 Identify whether transformations have been applied correctly	Data Consistency	Transform and Manipulate Data – pg. 47 N/A
Lesson 3 Video time - 00:05:23 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Identify Data Sets** Training and Test Sets Ensure Test Set is Representative	3.3 Identify training and test datasets 3.3.1 Separate available data into training and test datasets 3.3.2 Ensure test dataset is representative	N/A	Training and Test Data Sets – pg. 49 N/A
Lesson 4 Video time - 00:06:11 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Document Data Decisions** Assumptions, Predicates, and Constraints Make Information Available Identify Relevant Documentation	3.4 Document data decisions 3.4.1 List assumptions, predicates, and constraints upon which design choices have been reasoned 3.4.2 Make this information available to regulators and end users who demand deep transparency 3.4.3 Identify situations in which particular types of documentation are relevant	N/A	Documentation – pg. 51 N/A
Post-Assessment Assessment time - 01:00:00	Feature Engineering: Post-Assessment			

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Domain 4 Lesson Plan

Domain 4 - AI Algorithms and Models [approximately 4 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	AI Algorithms and Models: Pre-Assessment			
Lesson 1 Video time - 00:06:13 Exercise Lab time - 00:00:00 Workbook time - 00:10:00	**Consider Algorithm Applicability** Evaluate AI Algorithm Families NLP and Regression Decide on Suitable Algorithms	4.1 Consider applicability of specific algorithms 4.1.1 Evaluate AI algorithm families: 4.1.1.1 Supervised learning 4.1.1.2 Unsupervised learning 4.1.1.3 Reinforcement learning 4.1.1.4 Natural language processing (NLP) 4.1.1.5 Regression 4.1.2 Decide which algorithms are suitable, e.g., neural network, decision tree, k means	N/A	AI Algorithm Families – pg. 53 N/A Decide on Suitable Algorithms – pg. 54 N/A
Lesson 2 Video time - 00:04:50 Exercise Lab time - 00:00:00 Workbook time - 00:10:00	**Train a Model** Train with Parameters Tune by Changing Parameters Gather Performance Metrics Iterate As Needed	4.2 Train a model using the selected algorithms 4.2.1 Train model using an algorithm with starting parameters 4.2.2 Tune the model by changing parameters 4.2.3 Identify performance metrics for the model 4.2.4 Determine whether additional iterations are required	N/A	Train with Parameters – pg. 56 N/A Train a Model – pg. 57 N/A
Lesson 3 Video time - 00:06:02 Exercise Lab time - 00:00:00 Workbook time - 00:25:00	**Evaluate Model Performance** Check for Overfitting and Underfitting Generate Metrics or KPIs Introduce New Test Data	4.3 Evaluate model accuracy, precision, and sensitivity 4.3.1 Check for overfitting and underfitting 4.3.2 Generate metrics or KPIs (accuracy, precision, recall, F1 score, etc.) 4.3.3 Introduce new test data to cross-validate robustness, testing how model handles unforeseen data	N/A	Overfitting and Underfitting – pg. 59 N/A Generate Metrics or KPIs – pg. 60 N/A Introduce New Test Data – pg. 61 N/A
Lesson 4 Video time - 00:06:54 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Evaluate Model Sensitivity** Test for Model Sensitivity Test for Model Specificity Model Performance Improvement Techniques	4.3 Evaluate model accuracy, precision, and sensitivity 4.3.4 Interpret performance metrics (sensitivity, specificity, confusion matrix) 4.3.5 Identify appropriate techniques to improve model performance	N/A	Test for Model Sensitivity and Specificity – pg. 63 N/A Model Improvement Techniques – pg. 64 N/A
Lesson 5 Video time - 00:09:14 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**Look for Potential Bias** Verify Inputs Resemble Training Data Confirm Correlations Check for Data Imbalances Guard Against Self-Fulfilling Prophecies	4.4 Identify potential sources of bias 4.4.1 Verify that inputs resemble training data 4.4.2 Confirm that training data do not contain irrelevant correlations we do not want the classifier to rely on 4.4.3 Check for imbalances in data 4.4.4 Guard against creating self-fulfilling prophecies based upon historical biases	Understanding Correlation	Verify Inputs Resemble Training Data – pg. 66 N/A Irrelevant Correlations and Data Values – pg. 67 362-boaters.xlsx Guard Against Self-Fulfilling Prophecies – pg. 68 N/A
Lesson 6 Video time - 00:03:23 Exercise Lab time - 00:00:00	**Select Specific Model** Cost, Speed, and Other Factors Determine if Model Meets Explainability	4.5 Select specific model after experimentation 4.5.1 Consider cost, speed, and other factors when evaluating models 4.5.2 Determine whether the selected model meets explainability requirements	N/A	Cost, Speed, Explainability, and Other Factors – pg. 70 N/A

Domain 4 - AI Algorithms and Models [approximately 4 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Workbook time - 00:05:00				
Lesson 7 Video time - 00:06:18 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**Tell Data Stories** Identify Visualization Types Create Visualizations Look for Trends Verify That the Visualization Is Useful	4.6 Tell data stories and obtain stakeholder approval 4.6.1 Identify which visualization types will represent the data story most appropriately 4.6.2 Where feasible, create visualizations of the results 4.6.3 Look for trends 4.6.4 Verify that visualizations are useful for making a decision	N/A	Identify Visualization Types – pg. 72 N/A Create Visualizations – pg. 73 N/A Look for Trends and Verify Visualizations – pg. 74 N/A
Lesson 8 Video time - 00:03:55 Exercise Lab time - 00:04:00 Workbook time - 00:05:00	**Obtain Stakeholder Approval** Collect Results and Benchmark Risks Hold Evaluation Sessions	4.6 Tell data stories and obtain stakeholder approval 4.6.5 Collect results and benchmark risks and hold sessions to evaluate the solution	N/A	Results, Risks, and Evaluation Sessions – pg. 76 N/A
Post-Assessment Assessment time - 01:00:00	AI Algorithms and Models: Post-Assessment			

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Domain 5 Lesson Plan

Domain 5 - Application Integration and Deployment [approximately 3.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Application Integration and Deployment: Pre-Assessment			
Lesson 1 Video time - 00:16:12 Exercise Lab time - 00:00:00 Workbook time - 00:30:00	**Design a Production Pipeline** Plan for CI/CD Create a Pipeline Find a Solution Build App Connections Test AI Accuracy Test AI Robustness Test AI Latency Test App to Fit Size of Use Case	5.1 Design a production pipeline, including application integration 5.1.1 Plan for continuous integration and deployment (CI/CD) 5.1.2 Design a pipeline that integrates with existing data stores and connects to the application 5.1.3 Build the connection between the AI and the application 5.1.4 Test accuracy, robustness, and latency of AI through application 5.1.5 Test application to fit the size of use case (e.g., in AI for mobile applications)	N/A	AI Model Pipeline – pg. 78 N/A Using a Pipeline – pg. 79 N/A Testing AI Models – pg. 80 N/A
Lesson 2 Video time - 00:17:14 Exercise Lab time - 00:00:00 Workbook time - 00:40:00	**Address Potential Challenges** Understand Types of Challenges Understand Challenge Indicators Understand Challenge Mitigations Document for Maintenance Train a Support Team Implement a Feedback Mechanism Implement Drift Detector Implement Ways to Gather New Data	5.2 Identify potential challenges of models in production and plan for AI solution support 5.2.1 Understand the types of challenges you are likely to encounter 5.2.2 Understand the indicators of challenges 5.2.3 Understand how each type of challenge could be mitigated 5.2.4 Document the functions within the AI solution to allow for maintenance (updates, fixing bugs, handling edge cases) 5.2.5 Train a support team 5.2.6 Implement a feedback mechanism 5.2.7 Implement drift detector 5.2.8 Implement ways to gather new data	N/A	AI Model Challenges – pg. 82 N/A Documentation Support – pg. 83 N/A Feedback Mechanisms – pg. 84 N/A Maintaining AI Solution – pg. 85 N/A
Lesson 3 Video time - 00:05:04 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Build a Security Plan** Access Levels and Permissions Infrastructure Security Assess Risk Models	5.3 Build a security plan 5.3.1 Consider internal access levels or permissions 5.3.2 Consider infrastructure security 5.3.3 Assess the risk of using a certain model or potential attack surfaces (e.g., adversarial attacks on real-time learning models)	N/A	Access Levels and Permissions – pg. 87 N/A Infrastructure Security and Risk Models – pg. 88 N/A
Lesson 4 Video time - 00:06:15 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Train Customers on Product** Inform on Model Limitations Inform of Intended Model Usage Share Documentation Manage Customer Expectations	5.4 Train customers on how to use product and what to expect from it 5.4.1 Inform users of model limitations 5.4.2 Inform users of intended model usage 5.4.3 Share documentation 5.4.4 Manage customer expectations	N/A	Train Users – pg. 90 N/A
Post-Assessment Assessment time - 01:00:00	Application Integration and Deployment: Post-Assessment			

Domain 6 Lesson Plan

Domain 6 - Maintaining and Monitoring AI [approximately 2.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Maintaining and Monitoring AI: Pre-Assessment			
Lesson 1 Video time - 00:11:22 Exercise Lab time - 00:00:00 Workbook time - 00:25:00	**Engage in Oversight** Log App and Model Performance Use Robust Monitoring Systems Act Upon Alerts Drift and Degraded Mode Detect Failure to Support New Info Evaluate Outputs	6.1 Engage in oversight 6.1.1 Log and monitor application and model performance to facilitate security, debug, accountability, and audit 6.1.2 Act upon alerts 6.1.3 Observe the system over time in a variety of contexts to check for drift or degraded modes of operation 6.1.4 Detect ways the system fails to support new information 6.1.5 Evaluate outputs according to thresholds defined in regulatory requirements	N/A	Monitor Performance – pg. 92 N/A Drift and Degraded Modes – pg. 93 N/A Failure Detection – pg. 94 N/A Evaluate Outputs and Document Results – pg. 95 N/A
Lesson 2 Video time - 00:05:35 Exercise Lab time - 00:00:00 Workbook time - 00:05:00	**Assess Business Impact** Track Impact Metrics Compare Previous to New Metrics Act on Unexpected Metrics	6.2 Assess business impact (key performance indicators) 6.2.1 Track impact metrics to determine whether the solution has solved the problem 6.2.2 Compare previous metrics with new metrics when changes are made 6.2.3 Act on unexpected metrics by finding problems and fixing them	N/A	Assessing Business Impact – pg. 97 N/A
Lesson 3 Video time - 00:05:33 Exercise Lab time - 00:00:00 Workbook time - 00:05:00	**Measure Impacts** Analyze Subgroup Impact Identify and Mitigate Issues Identify Optimization Opportunities	6.3 Measure impacts on individuals and communities 6.3.1 Identify impact on specific subgroups 6.3.2 Identify and mitigate issues 6.3.3 Identify opportunities for optimization	N/A	Individual and Community Impact – pg. 99 N/A
Lesson 4 Video time - 00:05:58 Exercise Lab time - 00:00:00 Workbook time - 00:05:00	**Handle User Feedback** Measure User Satisfaction Assess Whether Users Are Confused Incorporate Feedback into Future Versions	6.4 Handle feedback from users 6.4.1 Measure user satisfaction 6.4.2 Assess whether users are confused (e.g., do they understand what the AI is supposed to do for them?) 6.4.3 Incorporate feedback into future versions	N/A	User Feedback – pg. 101 N/A
Lesson 5 Video time - 00:05:55 Exercise Lab time - 00:00:00 Workbook time - 00:05:00	**Improve or Decommission** Combine Impact Observations Retrain, Continue, or Decommission Conclusion	6.5 Evaluate operation on a regular basis 6.5.1 Combine impact observations (e.g., business, community, technology trends) to assess AI value 6.5.2 Decide whether to retrain AI, continue to use AI as-is, or to decommission AI	N/A	Improve or Decommission – pg. 103 N/A
Post-Assessment Assessment time - 01:00:00	Maintaining and Monitoring AI: Post-Assessment			